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(54) **SYSTEM AND METHOD FOR RUN-ON
DETECTION**

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(2013.01); *E03D 13/005* (2013.01)

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(57) **ABSTRACT**

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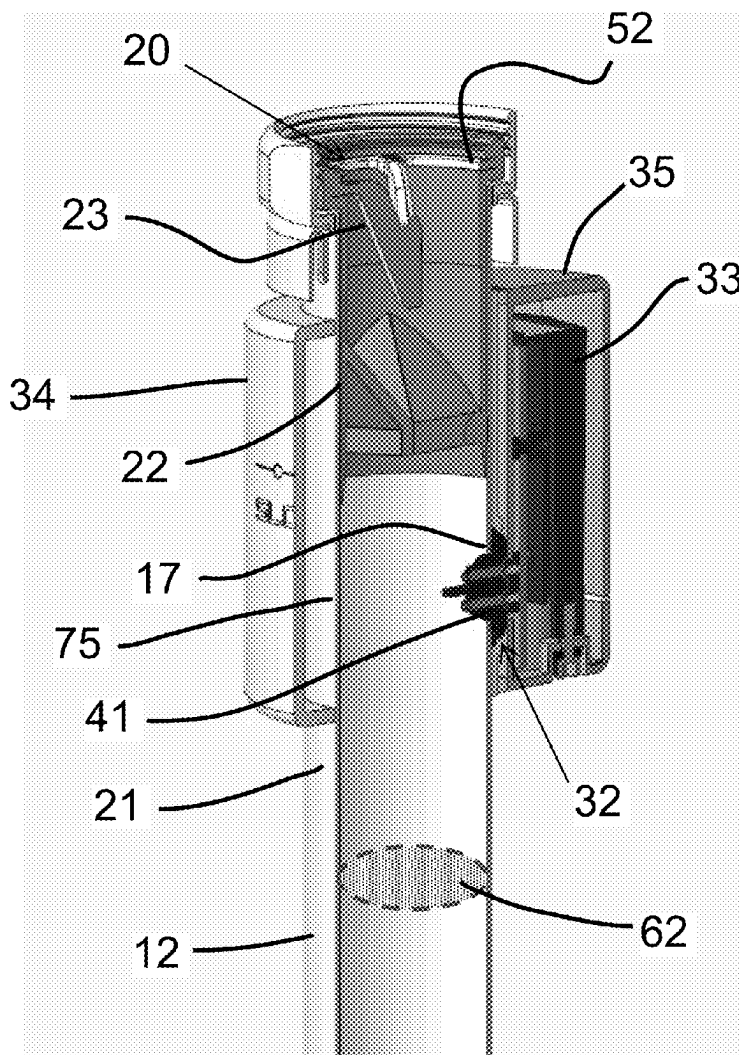
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A system for detection of water in a conduit includes a casing attached to the conduit on an exterior of the conduit, a sensor assembly supported by the casing and configured to sense a presence of water within the conduit, and an electronic component contained within the casing. The sensor assembly includes a first probe and a second probe, each having a probe contact accessible within the casing and extending into an interior of the conduit to be exposed in the interior of the conduit. The first probe and the second probe are spaced from each other and electrically isolated from each other. The electronic component is connected to the first probe contact and the second probe contact to place the electronic component in electronic communication with the first probe and the second probe. The conduit may be a vacuum breaker tube positioned between a valve and a plumbing fixture.



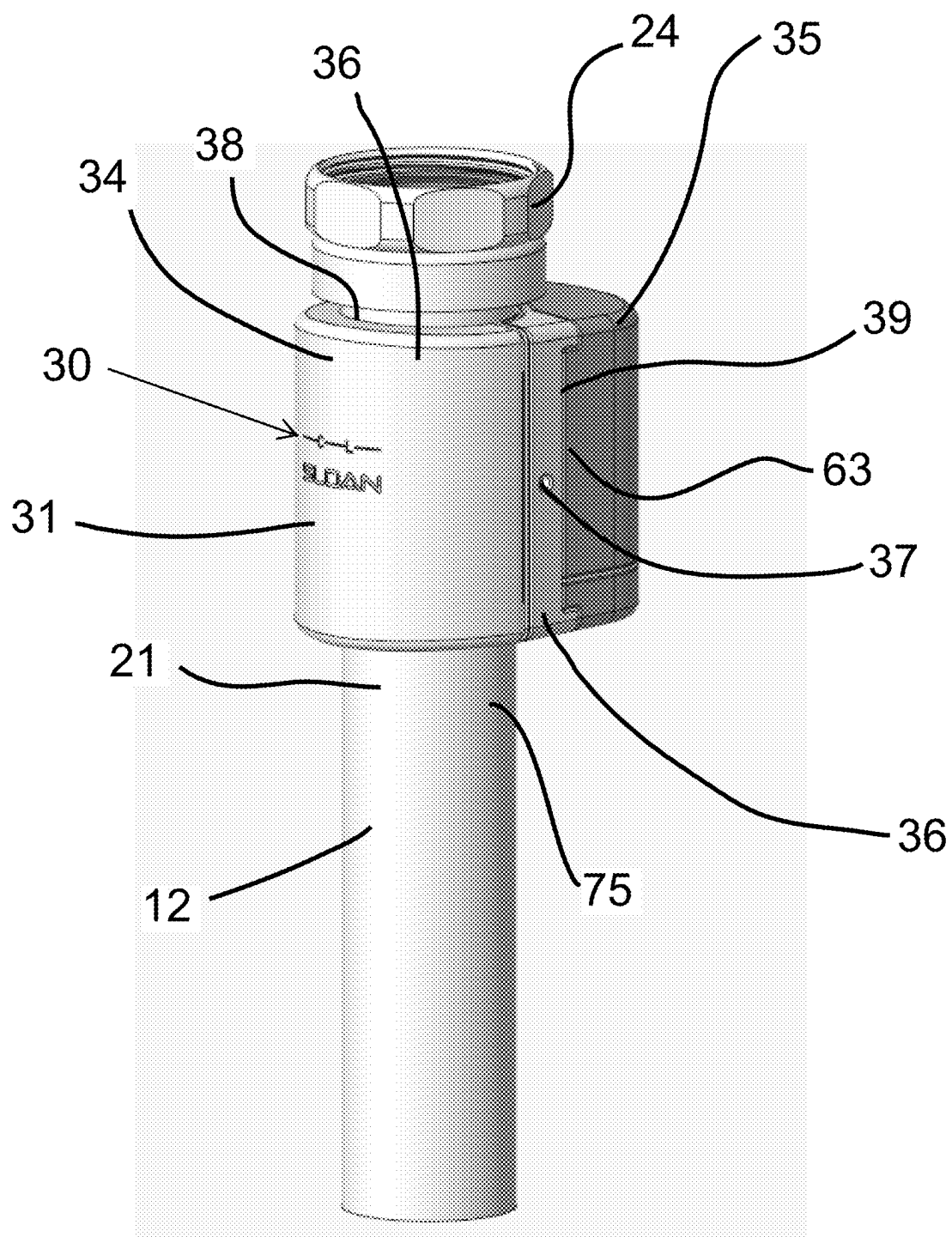


FIG. 1

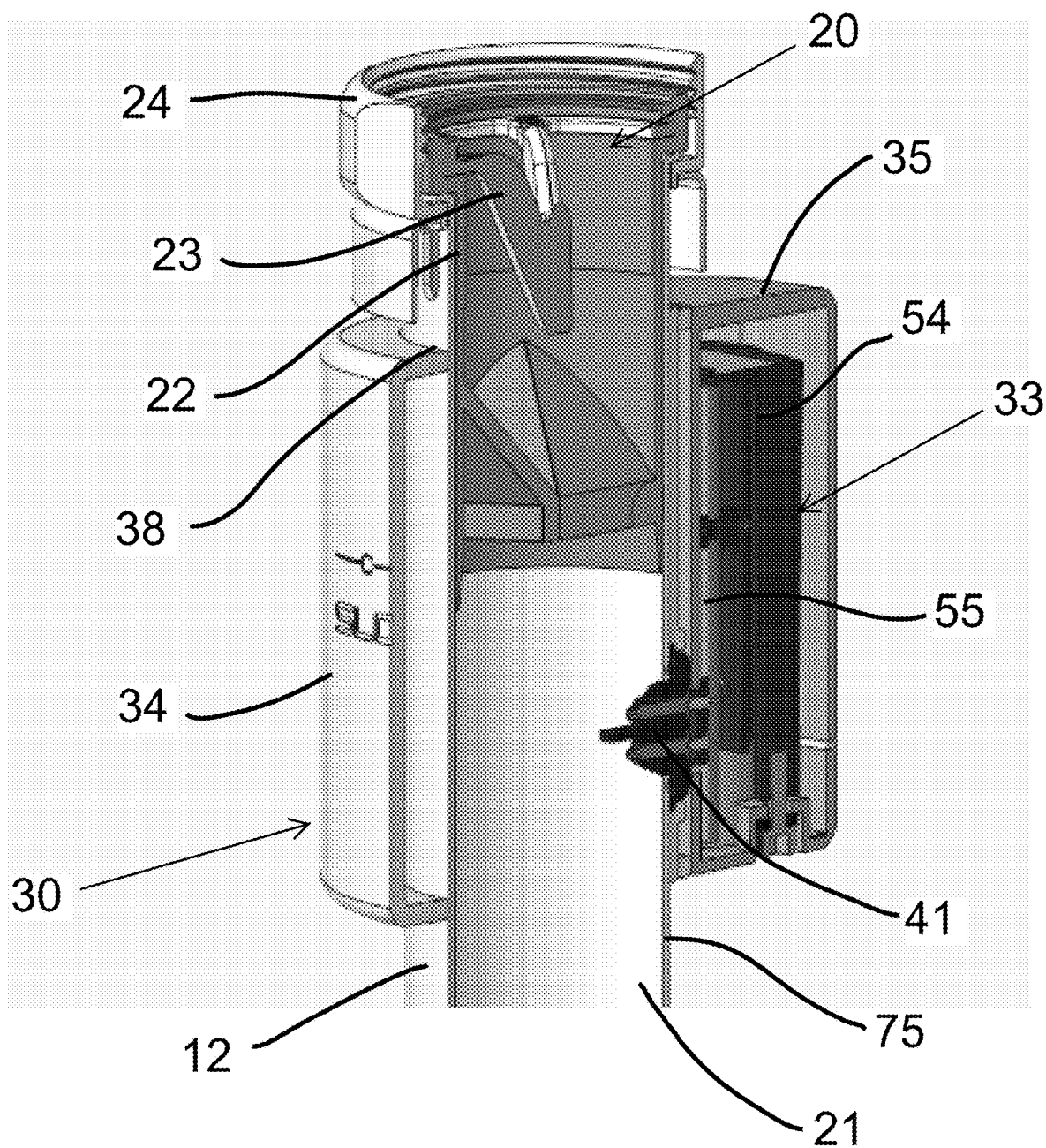


FIG. 3

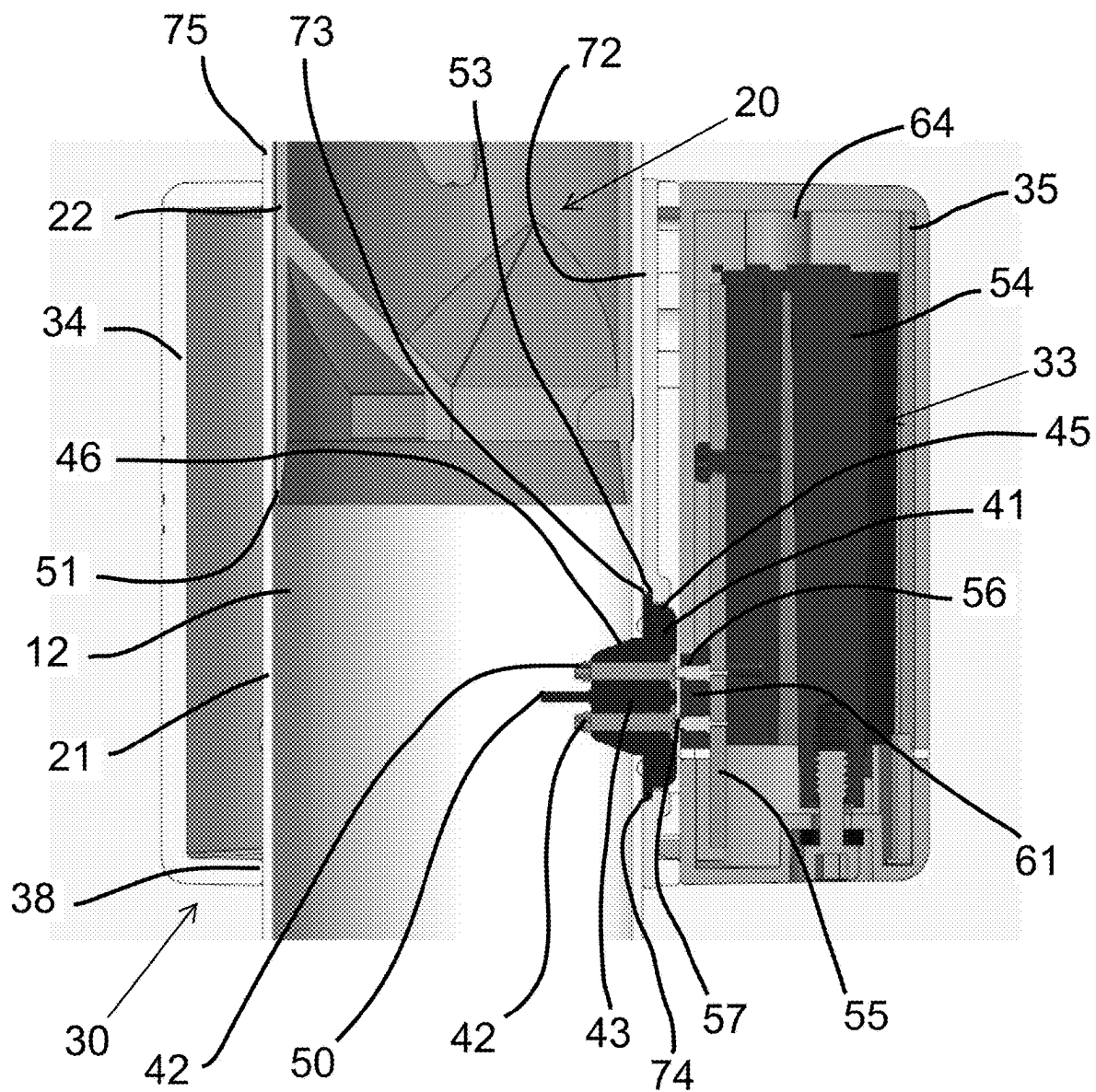


FIG. 4

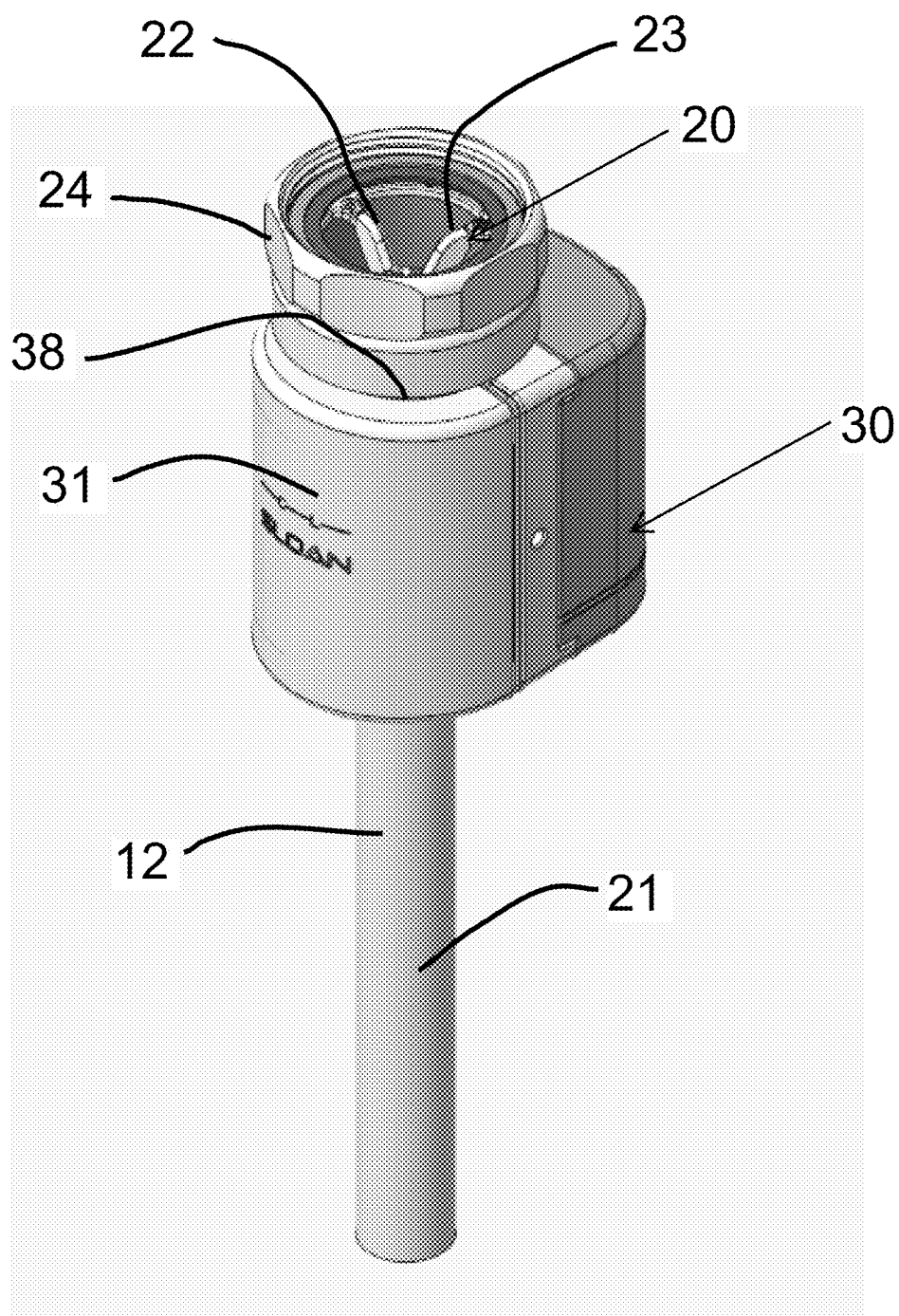
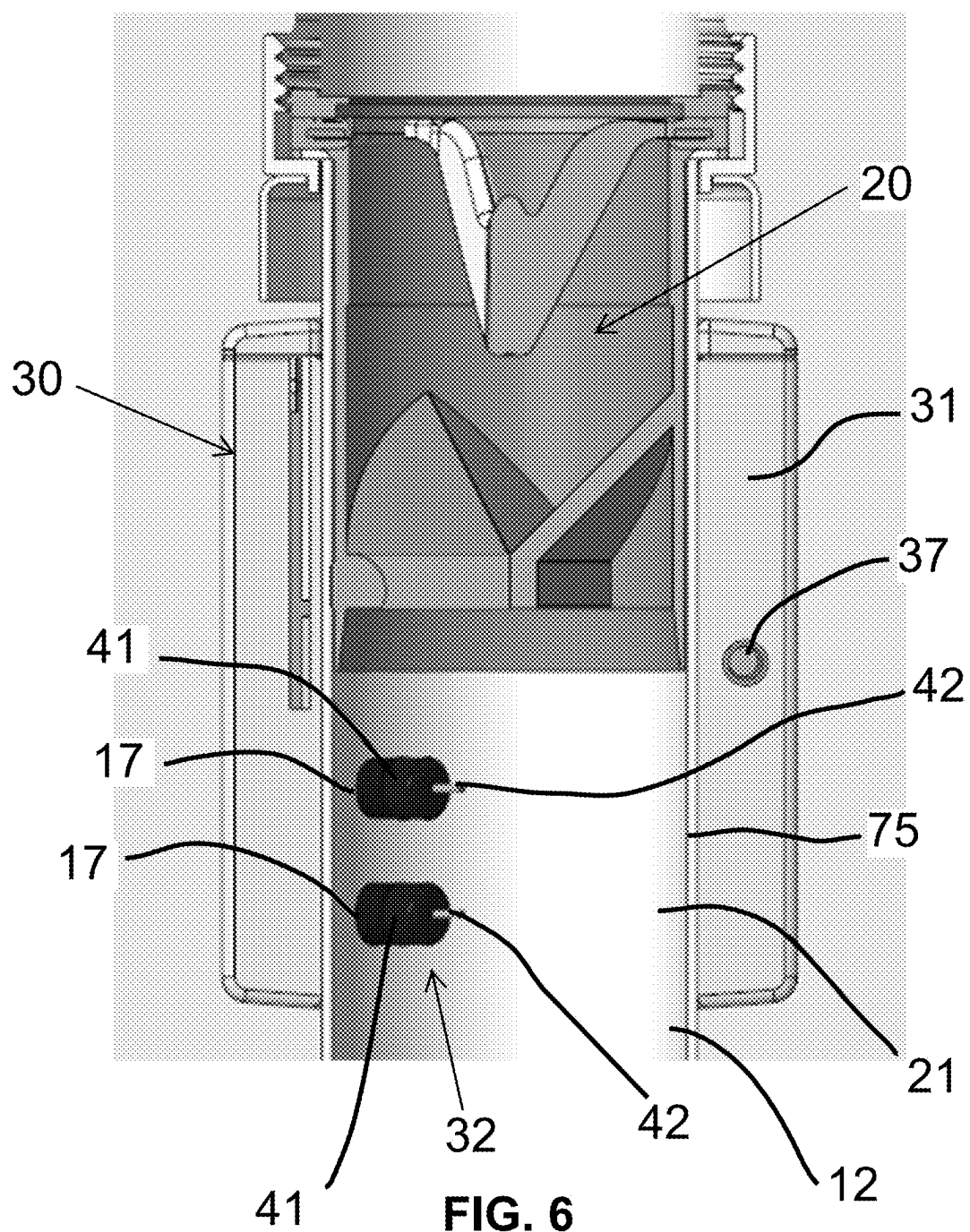
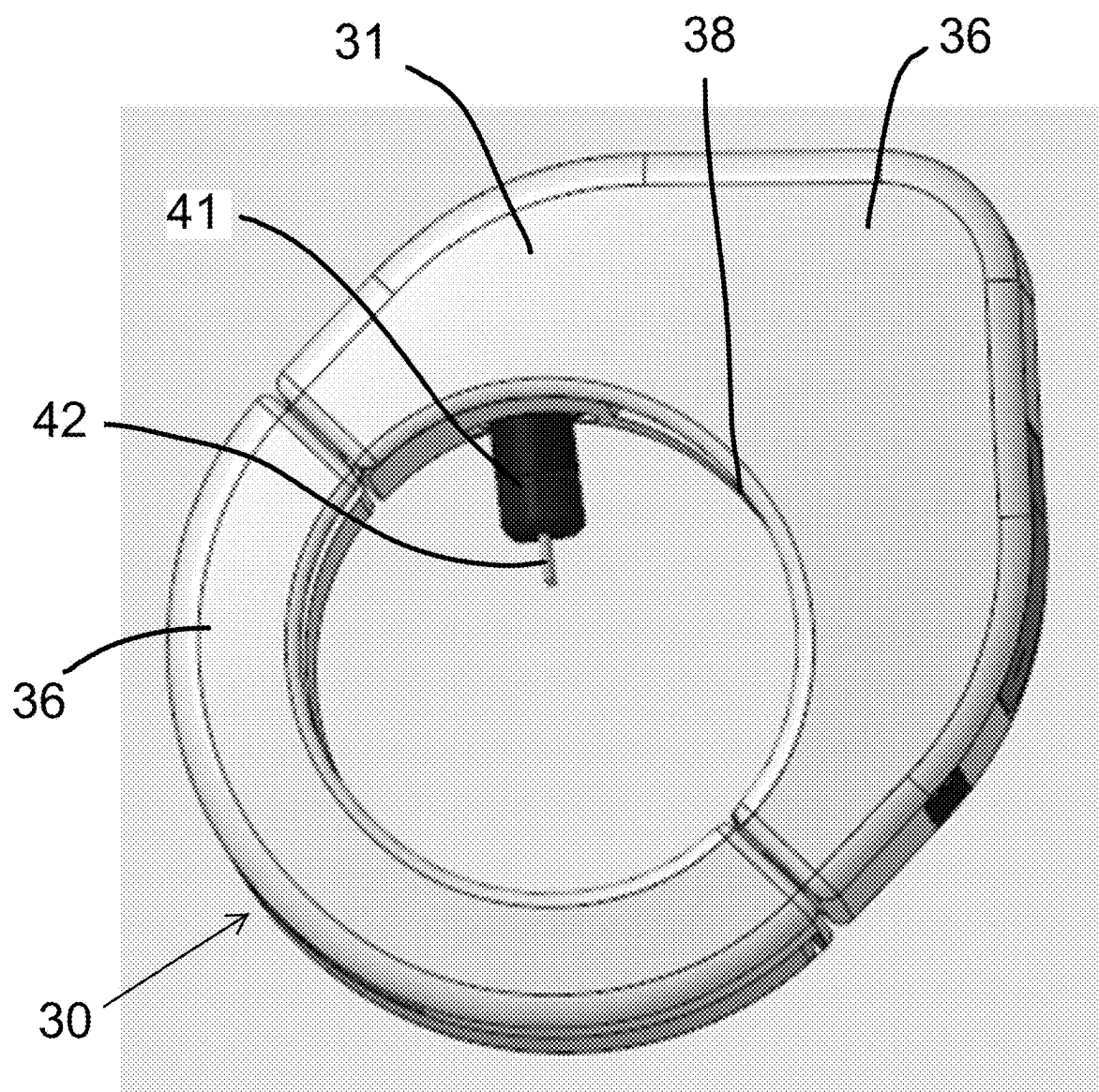


FIG. 5



**FIG. 7**

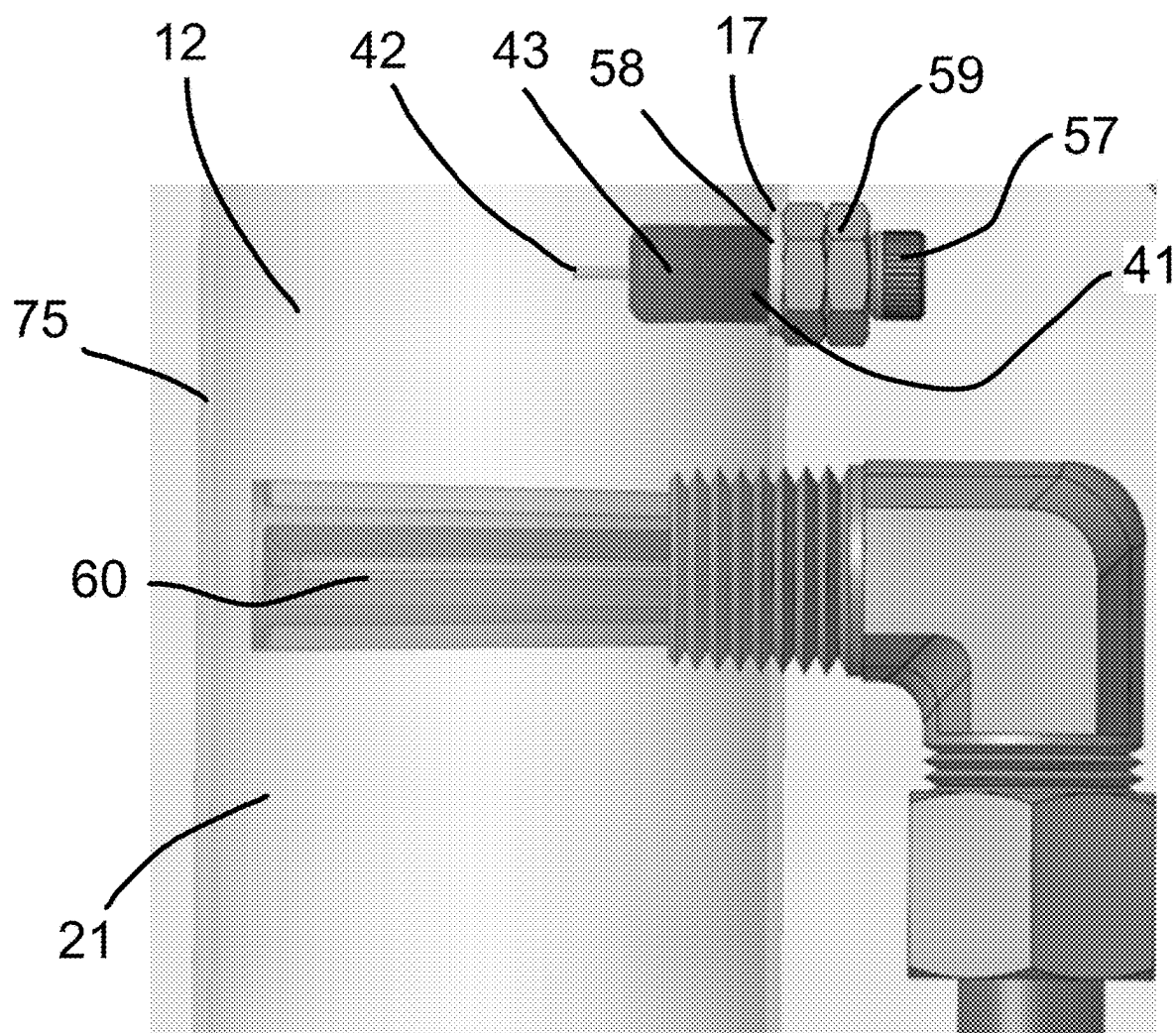
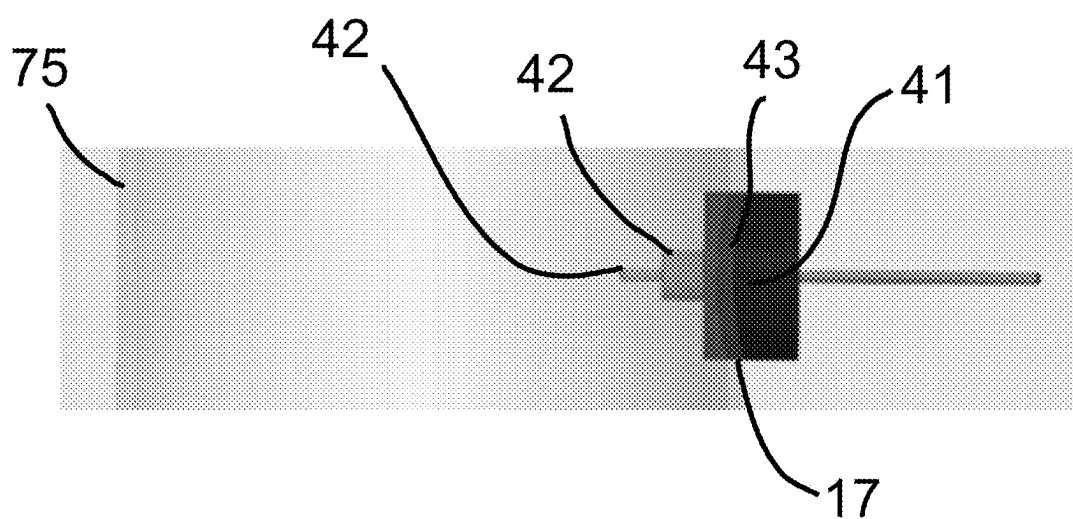
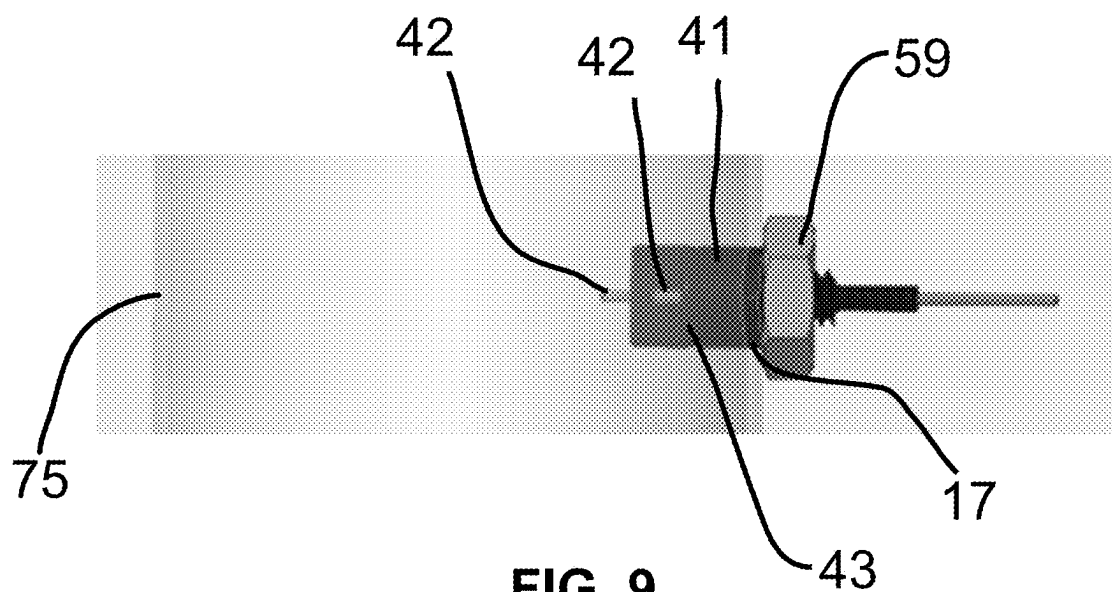


FIG. 8



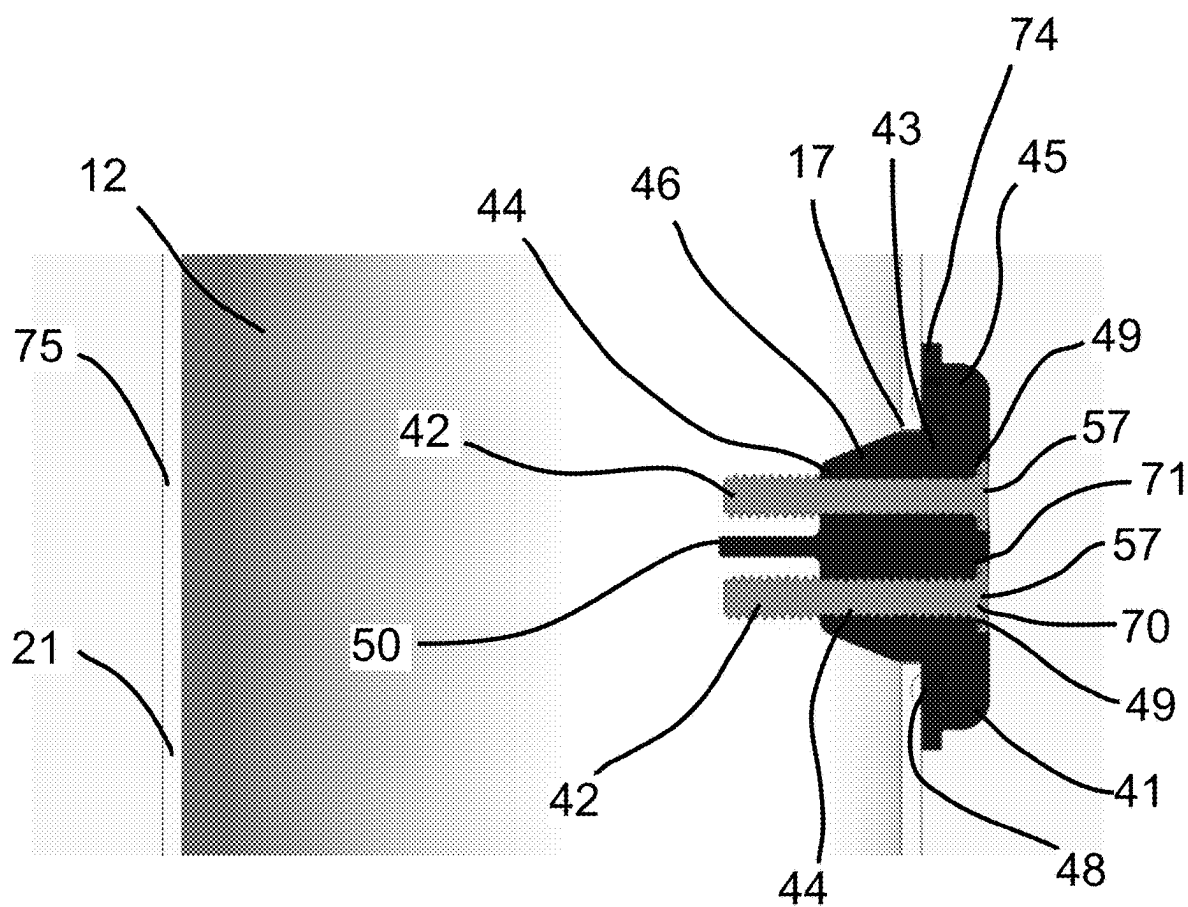


FIG. 11

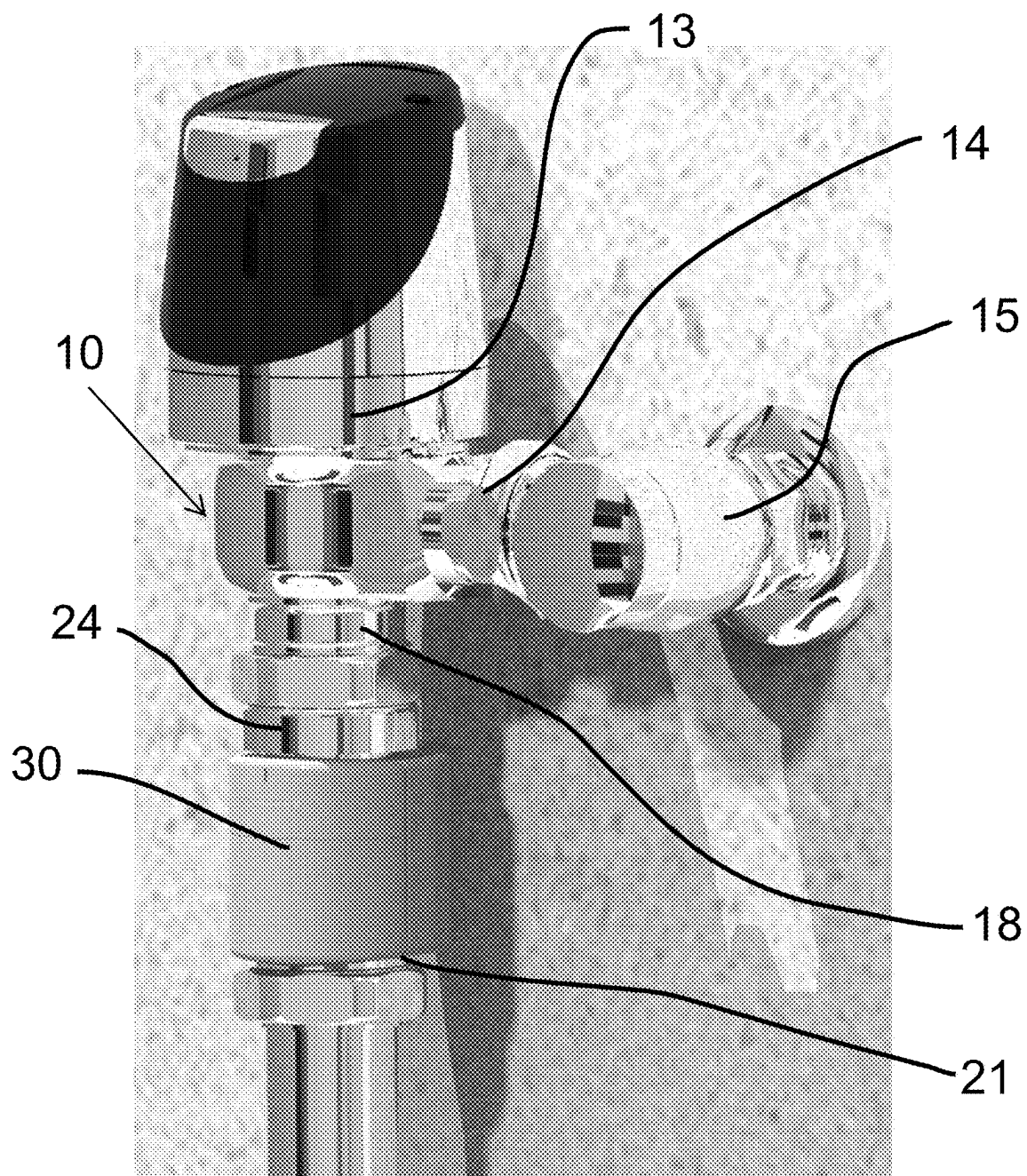


FIG. 12

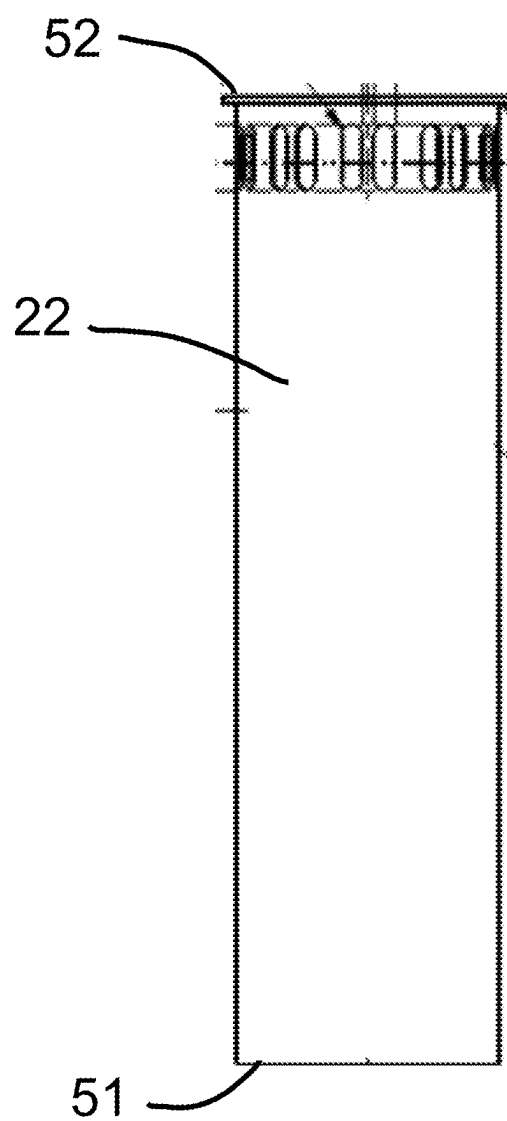


FIG. 13

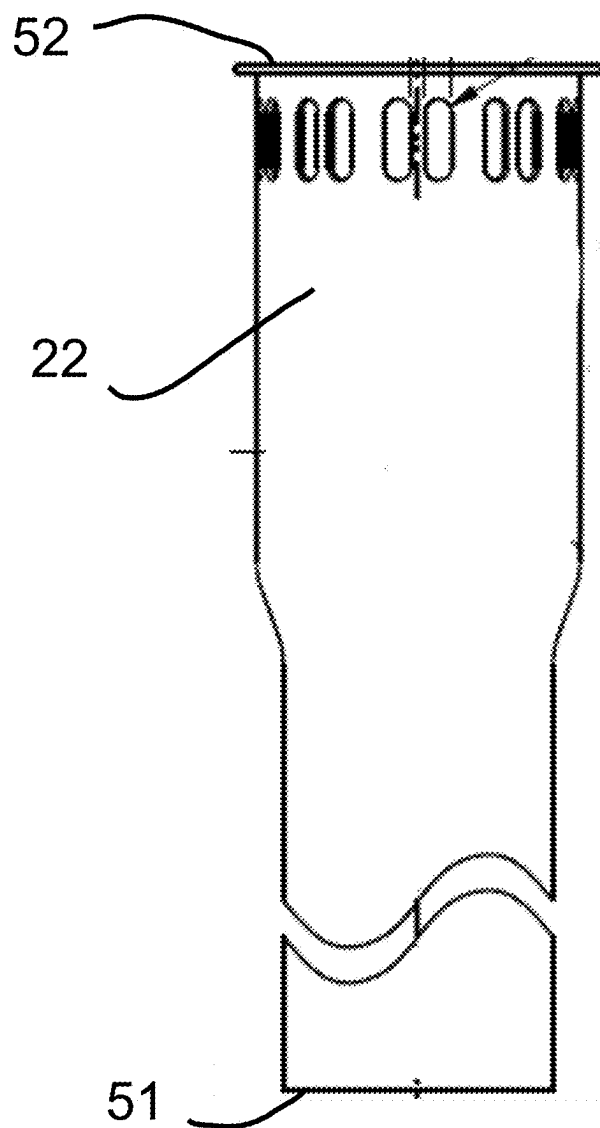


FIG. 14

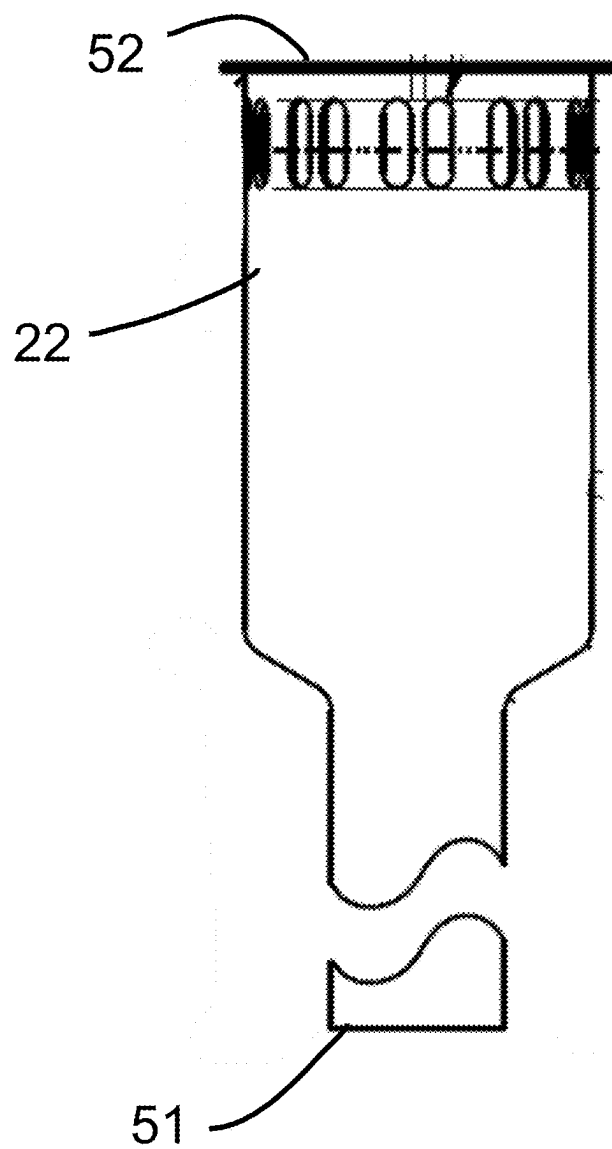


FIG. 15

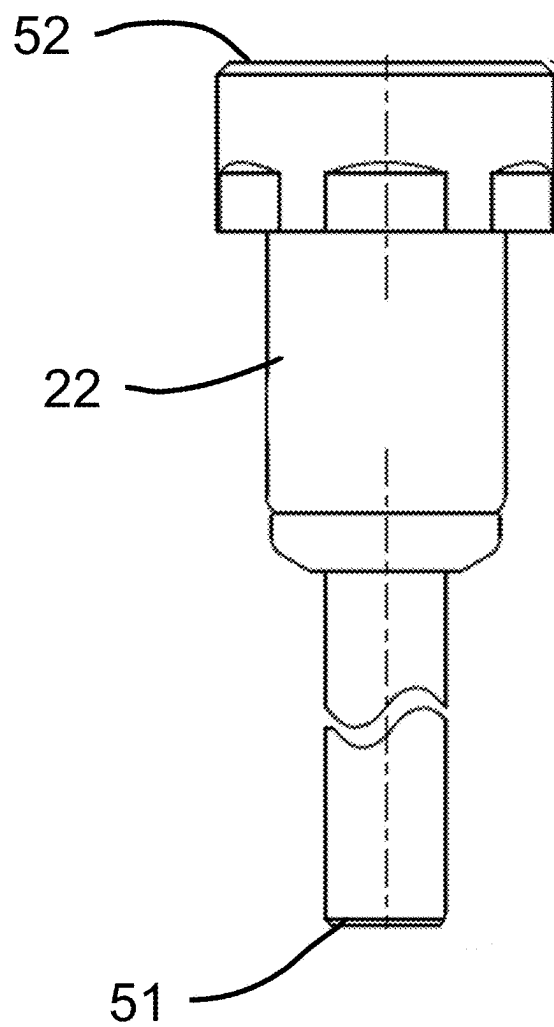


FIG. 16

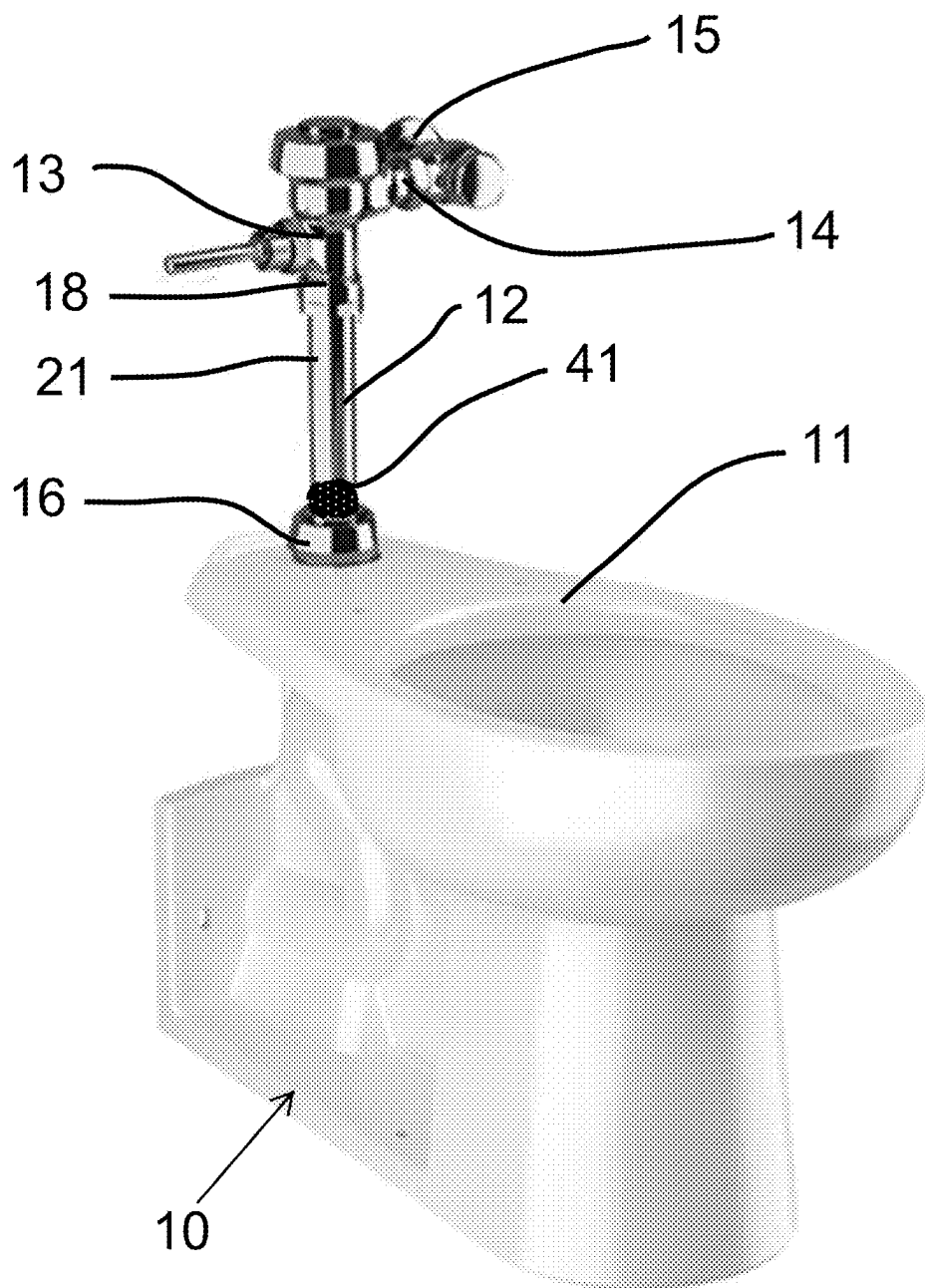
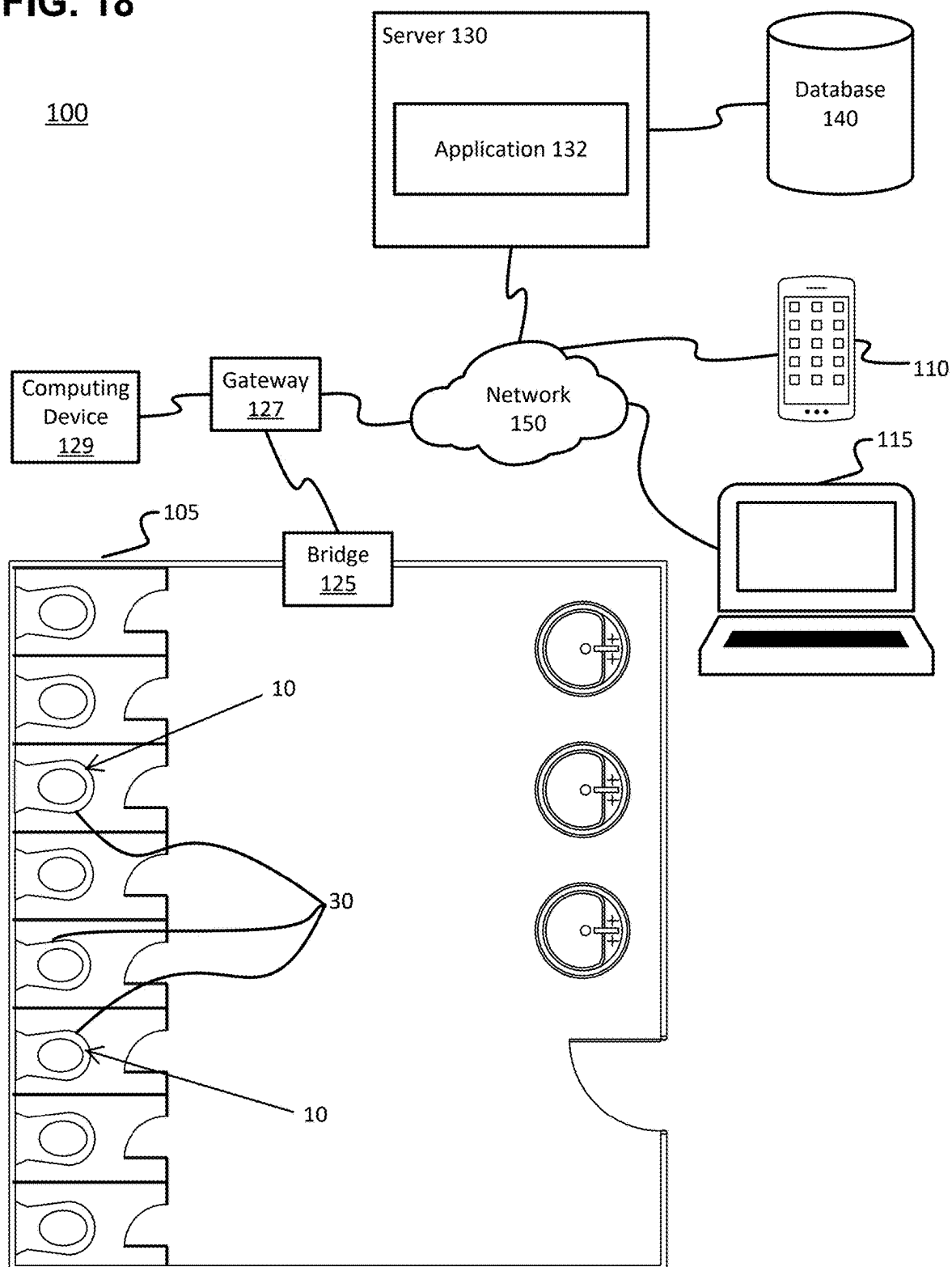


FIG. 17

FIG. 18



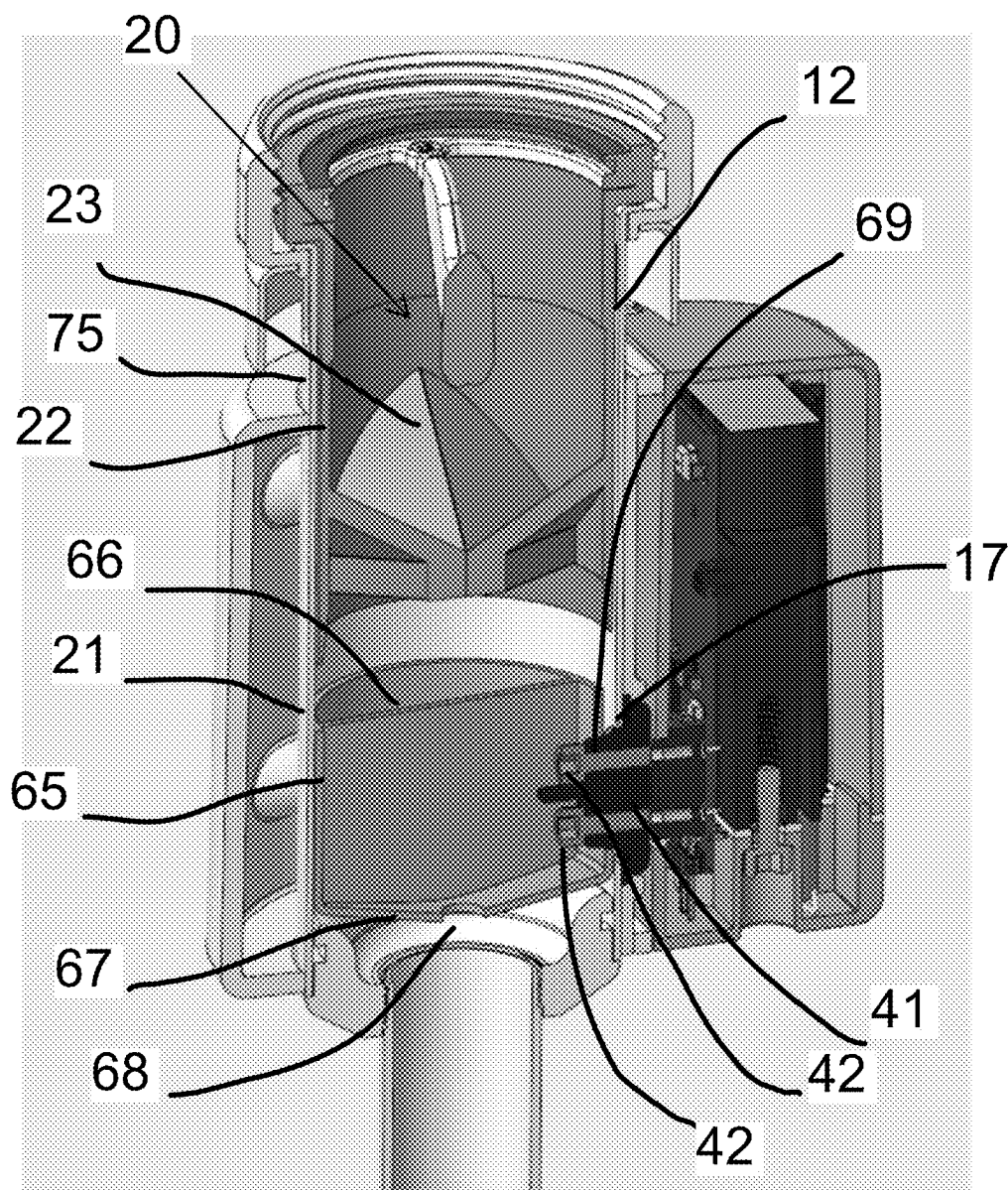


FIG. 19

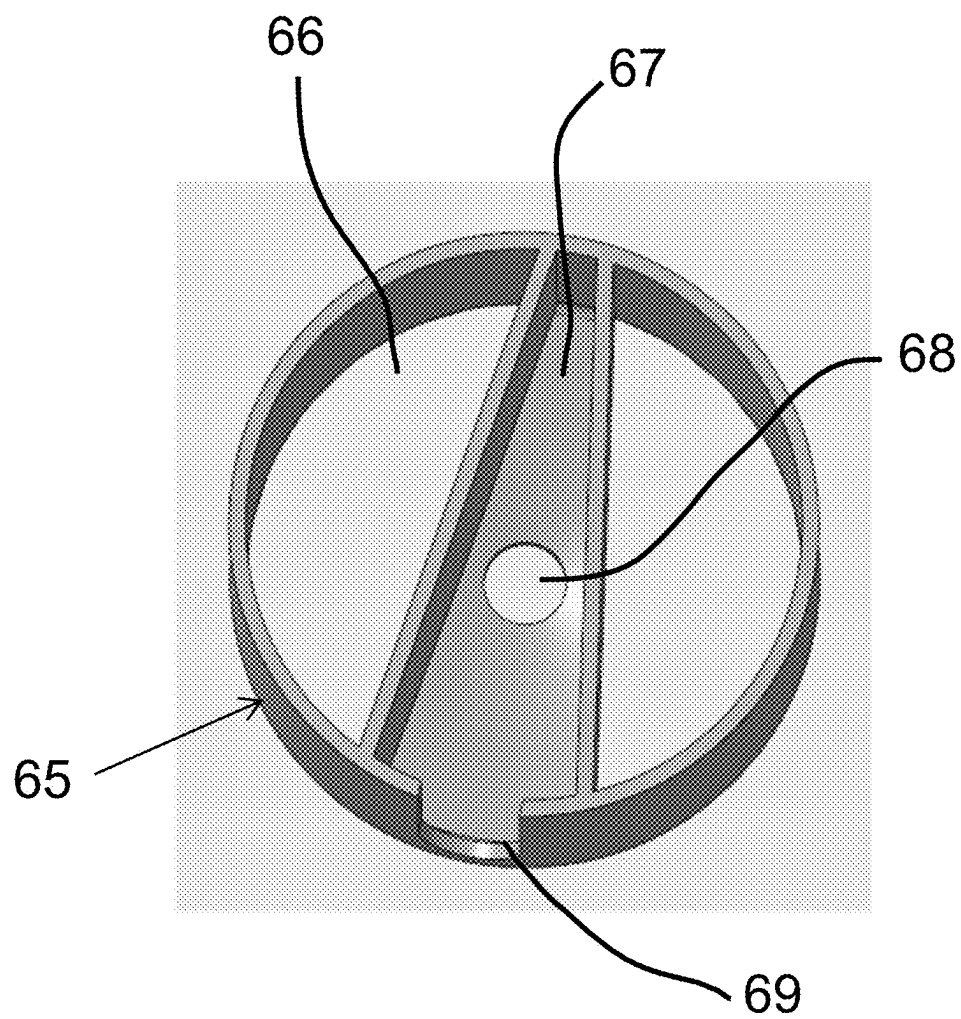


FIG. 20

SYSTEM AND METHOD FOR RUN-ON DETECTION

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a non-provisional of, and claims priority to, U.S. Provisional Application No. 63/556,509, filed Feb. 22, 2024, which prior application is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates to a system and method for detecting the presence of liquid in a pipe or other conduit, and more specifically to a system that includes a sensor placed to detect running of water from the outlet of a flushometer.

BACKGROUND

[0003] Plumbing fixtures of all types occasionally experience running of water at times when flow should be ceased. One particular example is a malfunction of a flushometer which causes water to run continuously through the flushometer (known as “run-on”). Run-on can be caused by various conditions, such as broken or damaged valve parts, debris within the valve, blockage of the valve bypass, insufficient water pressure, a malfunctioning actuator or solenoid, etc. This condition, combined with other conditions such as a clogged fixture, clogged drain line, and missing floor drain, can lead to extensive water damage in buildings. Additionally, as a flushometer flow rate can exceed 30 GPM, a run-on condition can lead to significant water waste, especially if unattended for an extended period such as a weekend or building closure. Run-on detection that is communicated to the appropriate personnel, which could be the maintenance team, building owner, etc., can help conserve water resources, reduce utility costs, and minimize the risk of water-related damage. Thus, electronic detection of run-on can be advantageous in enabling the issue to be addressed quickly. However, there is a need for a system for automated run-on detection that is effective and can be quickly and easily installed.

[0004] The present disclosure is provided to address this need and other needs in existing run-on detection devices and systems. A full discussion of the features and advantages of the present invention is deferred to the following detailed description, which proceeds with reference to the accompanying drawings.

BRIEF SUMMARY

[0005] General aspects of the present disclosure relate to systems and components for detecting the presence of water within a conduit, and in particular, to detect a low flow run-on condition within the conduit. The systems and components may utilize a sensor assembly capable of detecting water flowing within the conduit.

[0006] Aspects of the disclosure relate to a system for detection of water in a conduit including a casing configured to be attached to the conduit on an exterior of the conduit, a sensor assembly supported by the casing and configured to sense a presence of water within the conduit, and an electronic component contained within the casing. The sensor assembly includes a first probe having a first probe contact accessible within the casing and configured to extend into an

interior of the conduit to be exposed in the interior of the conduit, and a second probe having a second probe contact accessible within the casing and configured to extend into the interior of the conduit to be exposed in the interior of the conduit. The first probe and the second probe are spaced from each other and electrically isolated from each other. The electronic component is connected to the first probe contact and the second probe contact to place the electronic component in electronic communication with the first probe and the second probe.

[0007] According to one aspect, the casing includes a main casing configured to be attached to the conduit and supporting the sensor assembly, and a secondary casing removably connected to the main casing, wherein the secondary casing contains the electronic component. In one aspect, the system may also include a first contact and a second contact supported by the secondary casing and electronically connected to the electronic component, where the first contact and the second contact are positioned to engage the first probe contact and the second probe contact when the secondary casing is connected to the main casing, thereby placing the first probe and the second probe in electronic communication with the electronic component. In another aspect, the main casing may have a slot on an outer side thereof, and the secondary casing has a projection on an inner side thereof, and the secondary casing is removably connected to the main casing by the projection being received in the slot. In this configuration, the slot and the projection may be oriented horizontally, such that the secondary casing is removably connected to the main casing by the projection sliding horizontally into the slot. In a further aspect, the main casing has a central passage configured to receive the conduit therethrough, such that the main casing is configured to extend around the conduit, and the main casing includes two pieces joined together, with each of the two pieces defining a portion of the central passage.

[0008] According to another aspect, wherein the sensor assembly further includes a sensor body having a first passage and a second passage therethrough, the second passage spaced from the first passage, and the first probe and the second probe are received through the first passage and the second passage, respectively, and configured to be exposed in the interior of the conduit. The sensor body also includes an outer portion configured to be positioned on an exterior of the conduit and a projection configured to extend through a hole in the conduit and into an interior of the conduit. In one aspect, the outer portion may be wider than the projection and configured to permit sealing of the outer portion around the hole. In this configuration, the casing may have an inner wall configured to confront an exterior of the conduit and an opening in the inner wall receiving a portion of the outer portion of the sensor body, and the inner wall of the casing may be recessed around the opening to receive a lip of the outer portion of the sensor body, such that the lip of the sensor body is configured to be received between the inner wall of the casing and the exterior of the conduit.

[0009] According to a further aspect, the electronic component includes a power source configured to deliver power to the sensor assembly. The system may include a plurality of electronic components, including the electronic component, contained within the casing. The plurality of electronic components may include a power source, a processor, a

memory, and/or a communication device configured for communication with an external device.

[0010] Additional aspects of the disclosure relate to an assembly including a conduit configured to allow passage of water between a valve and a plumbing fixture and a system for detection of water in the conduit according to the aspects above. The assembly also includes a restrictor tray positioned within the interior of the conduit, adjacent to the first probe and the second probe. The restrictor tray includes a vessel configured to collect the water flowing through the conduit and to restrict a rate of the water exiting the vessel to a sufficient degree to permit the first probe and the second probe to contact the water during a low flow run-on condition.

[0011] Further aspects of the disclosure relate to a vacuum breaker assembly for use with a plumbing fixture and a valve configured to control discharge of water through an outlet and into the plumbing fixture. The vacuum breaker assembly includes a vacuum breaker tube configured to be connected in fluid communication with the outlet of the valve, between the valve and the plumbing fixture, the vacuum breaker tube having a wall and defining an interior to receive the discharge of the water from the outlet, a vacuum breaker sleeve received within the vacuum breaker tube and fixed in position within the vacuum breaker tube, and a vacuum breaker supported by the vacuum breaker sleeve. The vacuum breaker sleeve defines an inner passage, such that water discharged through the outlet is configured to flow through the vacuum breaker tube and the inner passage of the vacuum breaker sleeve to the plumbing fixture. The vacuum breaker is positioned within the inner passage of the vacuum breaker sleeve, and the vacuum breaker is configured to prevent backflow of water from the plumbing fixture into the valve. The assembly also includes a sensor assembly configured to sense a presence of water within the vacuum breaker tube and having a first probe extending through the wall of the vacuum breaker tube and into the interior of the vacuum breaker tube to be exposed in the interior of the vacuum breaker tube, and a second probe extending through the wall of the vacuum breaker tube and into the interior of the vacuum breaker tube to be exposed in the interior of the vacuum breaker tube. The first probe and the second probe are spaced from each other and electrically isolated from each other.

[0012] According to one aspect, the assembly includes a casing attached to the wall of the vacuum breaker tube and positioned on an exterior of the vacuum breaker tube, where the casing supports the first probe and the second probe of the sensor assembly. In one aspect, the assembly may also include an electronic component contained within the casing and connected to the first probe and the second probe to place the electronic component in electronic communication with the first probe and the second probe. In this configuration, the casing may include a main casing attached to the vacuum breaker tube and supporting the sensor assembly, and a secondary casing removably connected to the main casing, where the secondary casing contains the electronic component. In another aspect, the casing has a central passage configured to receive the vacuum breaker tube therethrough, such that the casing is configured to extend around the vacuum breaker tube.

[0013] According to another aspect, the sensor assembly further includes a sensor body having a first passage and a second passage therethrough, the second passage spaced

from the first passage, and the first probe and the second probe are received through the first passage and the second passage, respectively, and are exposed in the interior of the vacuum breaker tube. The sensor body also includes an outer portion positioned on an exterior of the wall of the vacuum breaker tube and a projection extending through a hole in the wall of the vacuum breaker tube and into the interior of the vacuum breaker tube.

[0014] According to a further aspect, a restrictor tray is positioned within the interior of the vacuum breaker tube, adjacent to the first probe and the second probe. The restrictor tray includes a vessel configured to collect the water flowing through the vacuum breaker tube and to restrict a rate of the water exiting the vessel to a sufficient degree to permit the first probe and the second probe to contact the water during a low flow run-on condition. In one aspect, the restrictor tray may include a drain in a bottom of the vessel, the drain configured to restrict the rate of the water exiting the vessel.

[0015] Still further aspects of the disclosure relate to a sensor for detecting presence of water in a conduit, including a sensor body having a first passage and a second passage therethrough, the second passage spaced from the first passage, a first probe received in the first passage and having a first probe contact accessible at the outer portion of the sensor body, the first probe extending through the first passage and configured to be exposed in the interior of the conduit, and a second probe received in the second passage and having a second probe contact accessible at the outer portion of the sensor body, the second probe extending through the second passage and configured to be exposed in the interior of the conduit. The sensor body also includes an outer portion configured to be positioned on an exterior of the conduit and a projection configured to extend through a hole in the conduit and into an interior of the conduit, and the outer portion is wider than the projection to permit sealing of the outer portion around the hole. The sensor body is formed of or contains an insulative material located between the first passage and the second passage, and the insulative material of the sensor body electrically isolates the first probe from the second probe when the first and second probes are received in the first and second passages, respectively.

[0016] According to one aspect, the sensor further includes a droplet shield extending from the projection of the sensor body between the first passage and the second passage. The droplet shield is configured to avoid a space between the first and second probes being bridged by a droplet of water within the conduit.

[0017] According to another aspect, the sensor further includes seals located between the first probe and the sensor body and between the second probe and the sensor body to resist passage of water between the sensor body and the first and second probes. In one aspect, the seals may be located within the first passage and the second passage.

[0018] According to a further aspect, the first probe has a first enlarged head, and the first passage has a first recess in the outer portion that receives the first enlarged head, and the second probe has a second enlarged head, and the second passage has a second recess in the outer portion that receives the second enlarged head. In one aspect, the sensor also includes first seals located between the first enlarged head and the first recess to resist passage of water between the sensor body and the first probe, and second seals located

between the second enlarged head and the second recess to resist passage of water between the sensor body and the second probe.

[0019] According to yet another aspect, the sensor further includes a face seal located on an inner surface of the outer portion of the sensor body and configured to engage an outer surface of the conduit around the hole to resist passage of water between the sensor body and the conduit.

[0020] Yet additional aspects of the disclosure relate to a system for detection of water in a conduit, including a casing configured to be attached to a wall of the conduit on an exterior of the conduit, the casing comprising a main casing configured to be attached to the conduit and having a central passage receiving the conduit therethrough, and a secondary casing removably connected to the main casing, a sensor assembly supported by the main casing, and an electronic component contained within the secondary casing. The sensor assembly is supported by the main casing in a position such that the sensor assembly is configured to be in engagement with the conduit and to extend through the wall of the conduit, and the sensor assembly configured to sense a presence of water within the conduit. The electronic component is connected to the sensor assembly to place the electronic component in electronic communication with the sensor assembly.

[0021] According to one aspect, the sensor assembly has a probe configured to extend through the wall of the conduit and having a probe contact exposed within the main casing, and the system further includes a contact supported by the secondary casing and electronically connected to the electronic component. The contact is positioned to engage the probe contact when the secondary casing is connected to the main casing, thereby placing the probe in electronic communication with the electronic component.

[0022] According to another aspect, the main casing has a slot on an outer side thereof, and the secondary casing has a projection on an inner side thereof, and the secondary casing is removably connected to the main casing by the projection being received in the slot. In one aspect, the slot and the projection are oriented horizontally, and the secondary casing is removably connected to the main casing by the projection sliding horizontally into the slot.

[0023] According to a further aspect, the main casing includes two pieces joined together, with each of the two pieces extending around a portion of the central passage, such that the two pieces combine to define the central passage. In one aspect, the secondary casing is removably connected to one of the two pieces of the main casing.

[0024] Other aspects of the disclosure relate to a method for detection of water in a conduit, and in particular, detection of a low flow run-on condition in the conduit. Aspects of this method may use the systems, assemblies, sensors, and/or other components described herein. The method includes determining whether a run-on condition exists in the conduit, which determination may be done by the electronic components of the system or assembly (i.e., the electronic components contained in the casing), an external computing device, or a combination of such components and devices. If the determination is performed by the detection system (e.g., the electronic components thereof), the system may also be configured to transmit a run-on detection message to one or more external computing devices, alerting the presence of a potential run-on condition. If the determination is performed by an external device,

the system may transmit data gathered by the sensor to the external device for analysis, and the external device may generate a run-on detection message if a potential run-on condition is determined. In either case, determination of a run-on condition may require more than a single detection of water in the conduit by the sensor. For example, the determination may incorporate the duration or frequency of water detection, and data regarding the operation condition of an upstream valve when the water is detected, among other criteria.

[0025] Other features and advantages of the disclosure will be apparent from the following description taken in conjunction with the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] To allow for a more full understanding of the present disclosure, it will now be described by way of example, with reference to the accompanying drawings in which:

[0027] FIG. 1 is a perspective view of one embodiment of an assembly including a plumbing conduit in the form of a vacuum breaker tube for a water closet and a detection system according to aspects of the disclosure;

[0028] FIG. 2 is a perspective cross-section view of the assembly of FIG. 1;

[0029] FIG. 3 is a magnified view of a portion of the assembly of FIG. 2;

[0030] FIG. 4 is a plan cross-section view of the assembly of FIG. 1;

[0031] FIG. 5 is a perspective view of another embodiment of an assembly including a plumbing conduit in the form of a vacuum breaker tube for a urinal and a detection system according to aspects of the disclosure;

[0032] FIG. 6 is a plan cross-section view of another embodiment of an assembly including a plumbing conduit and a detection system according to aspects of the disclosure;

[0033] FIG. 7 is a top view of the detection system of FIG. 6;

[0034] FIG. 8 is a side view of a plumbing conduit with a trap primer and another embodiment of a sensor according to aspects of the disclosure, with the plumbing conduit shown partially transparent to illustrate internal detail;

[0035] FIG. 9 is a side view of a plumbing conduit with another embodiment of a sensor according to aspects of the disclosure, with the plumbing conduit shown partially transparent to illustrate internal detail;

[0036] FIG. 10 is a side view of a plumbing conduit with another embodiment of a sensor according to aspects of the disclosure, with the plumbing conduit shown partially transparent to illustrate internal detail;

[0037] FIG. 11 is a side view of a plumbing conduit with a sensor as shown in FIGS. 1-4, with the plumbing conduit shown partially transparent to illustrate internal detail;

[0038] FIG. 12 is a perspective view of a portion of a plumbing fixture with another embodiment of an assembly including a plumbing conduit and a detection system according to aspects of the disclosure;

[0039] FIG. 13 is a perspective view of one example of a vacuum sleeve usable with embodiments of an assembly and a detection system according to aspects of the disclosure;

[0040] FIG. 14 is a perspective view of another example of a vacuum sleeve usable with embodiments of an assembly and a detection system according to aspects of the disclosure;

[0041] FIG. 15 is a perspective view of another example of a vacuum sleeve usable with embodiments of an assembly and a detection system according to aspects of the disclosure;

[0042] FIG. 16 is a perspective view of another example of a vacuum sleeve usable with embodiments of an assembly and a detection system according to aspects of the disclosure;

[0043] FIG. 17 is a perspective view of a plumbing fixture in the form of a water closet, with another embodiment of an assembly including a plumbing conduit in the form of a vacuum breaker tube for a water closet and a detection system according to aspects of the disclosure;

[0044] FIG. 18 show an example environment where a detection system may be implemented, according to aspects of the disclosure;

[0045] FIG. 19 is a perspective view of another embodiment of an assembly including a plumbing conduit in the form of a vacuum breaker tube for a urinal and a detection system according to aspects of the disclosure, with a local restrictor installed; and

[0046] FIG. 20 is a perspective view of the local restrictor of FIG. 19.

DETAILED DESCRIPTION

[0047] While this invention is susceptible of embodiments in many different forms, there are shown in the drawings and will herein be described in detail example embodiments of the invention with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the broad aspect of the invention to the embodiments illustrated. In the following description of various example structures according to the invention, reference is made to the accompanying drawings, which form a part hereof, and in which are shown by way of illustration various example devices, systems, and environments in which aspects of the invention may be practiced. It is to be understood that other specific arrangements of parts, example devices, systems, and environments may be utilized and structural and functional modifications may be made without departing from the scope of the present invention.

[0048] Referring to the figures, and initially to FIGS. 1-4, there is shown an example embodiment of a detection system 30 for detection of water in a conduit 12 (e.g., a plumbing conduit), which is usable with a plumbing fixture assembly 10. The plumbing fixture assembly 10 includes a plumbing fixture 11, which may be a toilet or urinal, as well as a conduit 12 connected to the fixture 11, a valve 13 located upstream of the conduit 12 to control delivery of water to the fixture 11 through an outlet 18, and an inlet 14 to provide water to the valve 13. The valve 13 may be a manual and/or automatic flushometer in one embodiment, and the inlet 14 may include a tail of the valve 13 and a control stop 15. FIG. 12 illustrates a valve 13 in the form of a combination manual and automatic flushometer, and FIG. 17 illustrates a plumbing fixture 11 in the form of a toilet or water closet, with a valve 13 in the form of a manual flushometer. Other forms of valves 13 and plumbing fixtures 11 may be used in other embodiments. The conduit 12 in the

embodiment of FIGS. 1-4 includes a vacuum breaker assembly 20 extending from the output of the valve 13 to the plumbing fixture 11, which may be connected to the fixture 11 at a spud 16 as shown in FIG. 17. The vacuum breaker assembly 20 generally includes a vacuum breaker tube 21, a vacuum breaker sleeve 22 received within the vacuum breaker tube 21, a vacuum breaker 23 supported within the vacuum breaker sleeve 22, and a vacuum breaker nut 24 for connection to the outlet 18 of the valve 13, as shown in FIGS. 12 and 17. The vacuum breaker 23 functions to prevent backflow of water from the fixture 11 into the valve 13. The system 30 in FIGS. 1-4 is configured for use with a conduit 12 in the form of a vacuum breaker tube 21 for a toilet or water closet fixture 11.

[0049] The system 30 includes a casing 31 configured for attachment to the conduit 12, a sensor assembly 32 supported by the casing 31 and extending into the conduit 12, and electronic components 33 contained within the casing 31 and electronically connected to the sensor assembly 32. The electronic components 33 may include a battery or other power source 54, a processor, a memory, a communication device (e.g., a wireless transmitter and/or receiver) for communication with an external device, an antenna 64, and/or any combination of such components, as well as other components known in the art. A printed circuit board (PCB) 55 connected to the power source 54 may contain one or more of the electronic components 33, such as the processor, the memory, the communication device, etc. The power source 54 may be configured to provide power to the other electronic components 33 and the sensor(s) 41. In one embodiment, the system 30 may further include a power generator (not shown), such as a solar panel or a turbine within the water flow path, which may be connected to the power source 54 to supply power. In another embodiment, the system 30 may not include a power source 54, and the electronic components 33 and the sensor(s) 41 may be powered by hardwired connection.

[0050] The casing 31 in the embodiment of FIGS. 1-4 includes a main casing 34 connected to the conduit 12 and a secondary casing 35 mounted on the main casing 34. In this configuration, the main casing 34 connects to the conduit 12 to hold the sensor assembly 32 in place and to mount the system 30 on the conduit 12, and the secondary casing 35 includes the electronic components 33, such that connecting the secondary casing 35 to the main casing 34 places the electronic components 33 in communication with the sensor assembly 32. This electronic connection is described in greater detail herein. The main casing 34 includes two pieces 36 that are connected together, e.g., by one or more fasteners 37, to define a central passage 38 for receiving the conduit 12. The two pieces 36 of the main casing 34 may be tightened around the conduit 12 by tightening the fastener(s) 37. The main casing 34 has a slot 39 on one end, and the secondary casing 35 has a projection 63 that fits in the slot 39 in a complementary arrangement, such that the projection 63 is inserted laterally into the slot 39 and the secondary casing 35 is slid laterally. The main casing 34 and/or the secondary casing 35 may include complementary retaining structures (not shown), such as locking tabs, fasteners, etc., to retain the secondary casing 35 to the main casing 34. These retaining structures may be disconnectable to allow removal of the secondary casing 35 for servicing the sensor assembly 32 and/or the electronic components 33, e.g., replacing a battery.

[0051] The sensor assembly 32 includes one or more sensors 41, each configured for detecting water in the conduit 12. A run-on condition will cause water to flow through the conduit 12, and may also cause the conduit 12 to fill with water due to back-pressure from the fixture 11. In one embodiment, the sensor 41 is a water contact sensor that includes two or more probes 42 in the form of electrodes. The probes 42 are spaced from each other such that water in the conduit 12 closes the circuit between the probes 42, enabling the presence of water to be detected. In the embodiment of FIGS. 1-4, the sensor 41 has two probes 42 that are spaced from each other by a distance that is large enough that the presence of water in the conduit 12 is necessary to close the circuit between the probes 42 (e.g., a single drop of water will not bridge the space) and small enough to fit within the conduit and ensure that water present in the conduit will close the circuit. For example, the space between the probes 42 may be between 0.0625 inch and 0.5 inch. In the embodiment of FIGS. 1-4, a single sensor 41 holds two probes 42 spaced from each other. In another embodiment, such as the embodiment of FIGS. 6-8, two sensors 41 are used, each of which holds a single probe 42. Additional sensors 41 and/or probes 42 may be used in other embodiments. In one embodiment, the probes 42 may be made from a conductive material that is also corrosion-resistant to prevent corrosion from changing the electrical properties of the probes 42, such as stainless steel. The sensor body 43 may be formed of a durable, electrically insulating material such as plastic or rubber, to electrically isolate the probes 42 from each other and from the conduit 12.

[0052] In further embodiments, different types of sensors 41 for detecting the presence of water in the conduit 12 may be used, including a pressure sensor, a flow sensor, an optical/infrared sensor, a capacitance sensor, an acoustic sensor, a temperature sensor, a magnetic (Hall effect) sensor, a turbine sensor, a strain gauge sensor, among others, or a combination of such sensors. It is understood that some such sensors may require modification of the casing 31, the positioning of the sensor 41, and/or other parts of the system 30.

[0053] The sensor 41 in FIGS. 1-4 is also shown separately in FIG. 11. This sensor 41 in this embodiment includes a sensor body 43 having two passages 44 therethrough, such that the probes 42 extend through the passages 44. The sensor body 43 has an outer portion 45 positioned on the exterior of the conduit 12, and a projection 46 extending through a hole 17 in the conduit 12 to the interior of the conduit 12. The outer portion 45 is wider than the projection 46 to permit sealing around the hole 17 and thereby resist leakage of water through the conduit 12, even when under pressure. In one embodiment, the hole 17 may be threaded, and the sensor body 43 may have threading to permit threaded insertion of the sensor 41. The outer portion 45 also has a face seal 48 to resist leakage. The sensor body 43 is made of an insulative material to protect the probes 42 and electrically insulate and/or isolate the probes 42 from each other to resist shorting. The probes 42 in this embodiment are in the form of bolts or screws that extend through the passages 44, and seals 49 (e.g., O-rings) are positioned within the passages 44 to resist leakage of water through the passages 44. The probes 42 configured as shown in FIG. 11 have enlarged heads 70, and the sensor body 43 has recesses 71 in the outer surface of the outer portion 45. The seals 49

are located between the inner surfaces of the enlarged heads 70 and the recesses 71. The enlarged head 70 of each probe 42 forms the probe contact 57 in this embodiment. The sensor 41 also has a droplet shield 50 extending inward into the conduit 12 from the projection 46, to avoid the space between the probes 42 being bridged by a droplet of water. The droplet shield 50 is in the form of an elongated finger in this embodiment. This sensor 41 is installed by forming a hole 17 through the wall 75 of the conduit 12 (i.e., the vacuum breaker tube 21) that is approximately the same size as the projection 46, such as by drilling or punching, and then pressing the sensor 41 through the hole 17.

[0054] In general, the sensor 41 is positioned between the valve 13 for which flow is detected and the fixture 11 to which the water is flowing. In one embodiment where a vacuum breaker assembly 20 is used between the valve 13 and the fixture 11, the sensor 41 is positioned on the vacuum breaker tube 21 below the vacuum breaker 23 and below the bottom 51 of the vacuum breaker sleeve 22. If the vacuum breaker tube 21 extends all the way to the connection of the conduit 12 to the fixture 11 (e.g., the spud 16), the sensor 41 may be placed at any location along the vacuum breaker tube 21. For example, in one embodiment, the sensor 41 may be placed at the spud 16, which is illustrated schematically in FIG. 17. In another embodiment, the vacuum breaker tube 21 may not extend the entire distance from the valve 13 to the fixture 11, and the sensor 41 may be placed in a different part of the conduit 12 below the end of the vacuum breaker tube 21. This positioning permits the sensor 41 to monitor the presence of water within the vacuum breaker tube 21 without interfering with the operation of the vacuum breaker assembly 20 and without requiring modification of the vacuum breaker assembly 20. Additionally, in some circumstances, positioning a sensor 41 above the bottom 51 of the vacuum breaker sleeve 22 may create interference with the operation of the sensor 41 due to air taken in by the vacuum breaker assembly 20. As illustrated in FIGS. 2-4, portions of the casing 31 and the electronic components 33 may be positioned above the bottom 51 of the vacuum breaker sleeve 22, with the sensor 41 positioned below the bottom 51. FIG. 12 illustrates a detection system 30 as disclosed herein connected to a vacuum breaker tube 21 below a valve 13 in the form of a combination manual and automatic flushometer and above a fixture (not shown) in the form of a toilet or water closet. In another embodiment, the sensor 41 may be positioned above the bottom 51 of the vacuum breaker sleeve 22 and below the top 52 of the vacuum breaker sleeve, to extend through both the vacuum breaker tube 21 and the vacuum breaker sleeve 22. Thus, in one configuration, the sensor 41 may be positioned at any location between the top 52 of the vacuum breaker sleeve 22 and the fixture 11.

[0055] In operation, the sensor 41 is installed by forming the hole 17 in the conduit 12 and inserting the projection 46 through the hole 17, and the main casing 34 is mounted on the conduit 12 over the sensor 41 as described herein. In another embodiment, the main casing 34 may be mounted on the conduit 12 before mounting the sensor 41 and/or forming the hole 17. The main casing 34 has an opening 53 that receives a portion of the outer portion 45 of the sensor body 43 therethrough, such that the passages 44 extend from the interior of the conduit 12 to the interior of the casing 31, and the ends of the probes 42 are exposed within the casing 31. The inner wall 72 of the main casing 34 may have a recess

73 around the opening 53 to provide close engagement with the sensor 41. The outer portion 45 of the sensor body 43 may have a lip 74 received in the recess 73, such that the lip 73 is received between the inner wall 72 and the exterior of the conduit 12. The secondary casing 35 is then connected to the main casing 34, to place the electronic components 33 in communication with the probes 42 of the sensor 41. In one embodiment, the system 30 includes two contacts 56 in the form of contact pins that are positioned to engage the probes 42 and are in communication with at least some of the electronic components 33, to permit gathering of data from the sensor 41. Each probe 42 in this embodiment has a probe contact 57 that is engaged by one of the contacts 56. The secondary casing 35 in the embodiment of FIGS. 1-4 completely contains all of the electronic components 33, and the secondary casing 35 includes a fitting 61 in the side wall that permits the contacts 56 to extend outside the secondary casing 35 to contact the sensor 41. The secondary casing 35 may be releasably sealed in one embodiment, to permit servicing of the internal components thereof. In one embodiment, the system 30 may be provided as part of an assembly including the vacuum breaker assembly 20 or the vacuum breaker tube 21 with the system 30 (or one or more components thereof, such as the sensor(s) 41 and/or the main casing 34) already installed.

[0056] FIG. 5 illustrates another embodiment of a system 30 for detection of water in a conduit 12, according to aspects of the disclosure. The system 30 in FIG. 5 is designed for use with a vacuum breaker tube 21 for a urinal. The vacuum breaker tube 21 in this embodiment is narrower than in the embodiment of FIGS. 1-4, and the central passage 38 of the casing 31 is therefore smaller than in the embodiment of FIGS. 1-4. The system 30 in FIG. 5 is otherwise similar or identical to the system 30 of FIGS. 1-4, and will not be described again in detail for the sake of brevity.

[0057] FIGS. 6-7 illustrate another embodiment of a system 30 for detection of water in a conduit 12 that includes many features in common with the embodiment of FIGS. 1-4 described herein. Such common components and features will not be described again in detail with respect to the embodiment of FIGS. 6-7, and similar reference numbers will be used in the description thereof. The system 30 of FIGS. 6-7 includes a sensor assembly 32 with two sensors 41, each with a single probe 42. The sensor 41 in this embodiment is illustrated in greater detail in FIG. 8. Each sensor 41 has a sensor body 43 with a passage 44 there-through for the probe 42, which is in the form of a thin wire. A seal 58 is positioned around the sensor body 43 at the hole 17 to avoid leakage, and two nuts 59 are threaded onto the outer end of the sensor body 43 to hold the sensor 41 tightly in engagement with the conduit 12. Each sensor 41 also has a sensor contact or probe contact 57 at the outer end that is in communication with the probe(s) 42. The casing 31 in the embodiment of FIGS. 6-7 does not have separate main and secondary casings as in the embodiments of FIGS. 1-5. The casing 31 includes two pieces 36 that are connected together, e.g., by one or more fasteners 37, to define a central passage 38 for receiving the conduit 12, similar to the construction of the main casing 34 in FIGS. 1-4. The two pieces 36 of the casing 31 in this embodiment may be tightened around the conduit 12 by tightening the fastener(s) 37, as similarly described with respect to the embodiment of FIGS. 1-4. While FIGS. 6-7 do not illustrate the electronic components

within the casing 31, it is understood that the casing may include electronic components as described herein, including at least two contacts (not shown) similar to the contacts 56 of the embodiment in FIGS. 1-4, for engaging the sensor contacts 57 of the sensors 41 to place the electronic components in communication with the probes 42.

[0058] FIG. 8 also illustrates the presence of a trap primer 60 mounted on the conduit 12 to dispense water into the conduit 12. In certain embodiments, the detection systems 30 disclosed herein are usable with a conduit 12 that includes a trap primer 60, as long as the position and configuration of the trap primer 60 is such that the trap primer 60 does not contact or otherwise interfere with the sensor(s) 41. The sensor(s) 41 may be mounted upstream or downstream of the trap primer 60, depending on the location and configuration of the trap primer 60.

[0059] FIGS. 9 and 10 illustrate other embodiments of sensors 41 with probes 42 in the form of electrodes that may be used in connection with a detection system 30 as disclosed herein, including any of the embodiments of FIGS. 1-7. FIGS. 9 and 10 both illustrate sensors 41 with two probes 42. FIG. 9 illustrates a dual-probe sensor 41 with a longer probe 42 and a shorter probe 42 separated by a space, and the sensor 41 has a sensor body 43 and a nut 59 threaded onto the outer end of the sensor body 43 to hold the sensor 41 tightly in engagement with the conduit 12. The sensor 41 of FIG. 9 may also include sensor contacts similar to the sensor contacts 57 of FIGS. 6-8 at the outer end that are in communication with the probes 42, in one embodiment. In another embodiment, the probes 42 may connect to the electronic components of the detection system 30 to place the electronic components in communication with the probes 42 using a different structure. FIG. 10 illustrates a coaxial sensor 41 with an outer probe 42 and an inner probe 42 that is coaxially disposed with respect to the outer probe 42. The probes 42 are spaced from each other, and may be separated by an insulative material as discussed herein. The sensor 41 has a sensor body 43 that extends through the hole 17 and is held tightly in engagement with the conduit 12. The sensor 41 of FIG. 10 may also include sensor contacts similar to the sensor contacts 57 of FIGS. 6-8 at the outer end that are in communication with the probes 42, in one embodiment. In another embodiment, the probes 42 may connect to the electronic components of the detection system 30 to place the electronic components in communication with the probes 42 using a different structure. In either of the embodiments of FIGS. 9 and 10, a seal (not shown) similar to the seal 58 of FIGS. 6-8 may be positioned around the sensor body 43 at the hole 17 to avoid leakage. The sensors 41 in FIGS. 6-10 may all be used with a threaded hole 17 and threading around the sensor body 43 in one embodiment. In a further embodiment, the sensor 41 may be configured to have electrical communication between the electronic components 33 and the conduit 12, so that the conduit 12 itself is used as a probe. Only a single additional probe 42 is necessary in such an embodiment.

[0060] FIGS. 13-16 illustrate different configurations of vacuum breaker sleeves 22 that may be usable in connection with a system 30 as disclosed herein. FIG. 13 illustrates a one-piece vacuum breaker sleeve 22 with a constant diameter of 1.5 inch. FIG. 14 illustrates a one-piece vacuum breaker sleeve 22 with a taper from a diameter of 1.5 inch to a diameter of 1.25 inch. FIG. 15 illustrates a one-piece vacuum breaker sleeve 22 with a taper from a diameter of

1.5 inch to a diameter of 0.75 inch. FIG. 16 illustrates a three-piece vacuum breaker sleeve 22 with an upper diameter of 1.5 inch and a lower diameter of 0.75 inch. It is understood that the break lines in the drawings indicate that a portion of the length of the vacuum breaker sleeve 22 is not shown. In the vacuum breaker sleeves 22 of FIGS. 14-16, the sensor 41 may be installed below the taper, at or below the smaller-diameter portion of the vacuum breaker sleeve 22, in one embodiment.

[0061] In some embodiments, the system 30 may be used for a plumbing fixture assembly 10 designed for a low-flow application, e.g., a high-efficiency urinal at low pressure. For an application of this type, the conduit 12 may not be completely filled with water in a run-on situation. In one embodiment, the geometry of the sensor 41 may be modified to detect water in a condition when the conduit 12 is not completely filled. In another embodiment, the system 30 may be used with a flow restrictor (variable or fixed) downstream from the sensor 41, to cause the conduit 12 to fill completely or to a greater level during a run-on condition. FIG. 2 schematically illustrates such a flow restrictor 62 within the conduit 12. In another embodiment, a local restrictor may be used to cause pooling of water around the sensor 41 while minimally affecting overall backpressure within the conduit 12. An example of a local restrictor 65 is shown in FIGS. 19-20, in the form of an insert mounted within the conduit 12. The local restrictor 65 in this embodiment is mounted within the conduit and has a cylindrical shape to define a central passage 66, with a restrictor tray 67 supported within the central passage 66 adjacent to the sensor 41. The restrictor tray 67 in this embodiment is in the form of a cup or other vessel that is open at the top, and is configured to restrict water exiting the restrictor tray 67 to a degree sufficient that water collects within the restrictor tray 67 during a low flow run-on condition to enable the probes 42 to contact the water, and the sensor 41 to thereby detect the water. In the embodiment of FIGS. 19-20, the restrictor tray 67 has a drain 68 that allows water to slowly drain from the restrictor tray 67, and the bottom of the restrictor tray 67 is sloped toward the drain 68 to ensure that water does not collect permanently within the tray 67. A different type of restriction structure may be used in place of the drain 68 in other embodiments. The local restrictor 65 further has a slot 69 in the side wall thereof, to enable the probe(s) 42 of the sensor 41 to extend into the restrictor tray 67. The slot 69 is positioned adjacent to the hole 17 when the local restrictor 65 is mounted within the conduit 12 just below the vacuum breaker sleeve 22. It is understood that the geometries of the slot 69 (or slots) and the restrictor tray 67 may be modified for embodiments where multiple sensors 41 and/or a different type of sensor 41 is used.

[0062] FIG. 18 shows an environment 100 where the detection system 30 described herein may be implemented. As shown in FIG. 18, the environment 100 comprises a restroom 105, a first user device 110, a second user device 115, and a server 130 interconnected via network 150.

[0063] Restroom 105 may be a bathroom in a commercial space, such as an office building, a retailer (e.g., mall), a stadium, etc. The restroom 105 may comprise a plurality of water closets and a plurality of sinks. Although not shown in FIG. 18, the restroom 105 may also comprise urinals, hand dryers, hand sanitation units, etc. (individually referred to as a “fixture,” and collectively as “fixtures”). Each fixture may comprise one or more sensors (not shown). Each detection

system 30 may allow a respective fixture to operate in a touchless, or a near touchless, manner. The fixture and/or the detection system 30 may be hardwired into a building’s electrical supply. Alternatively, the fixture and/or the detection system 30 may be battery-operated. In yet further examples, the fixtures and/or the one or more of sensors may receive power from a low-voltage power supply, such as power-over-ethernet (PoE) or from a transformer located in the restroom 105. The detection system 30 may be configured to transmit a run-on detection message to one or more computing devices, such as a local computing device 129, the first user device 110, the second user device 115, and/or the server 130, if water is detected in the conduit 12. Additionally or alternatively, the detection system 30 may include connections and/or logic to determine when the respective fixture has been activated (e.g., a toilet or urinal flushed), so that run-on can be determined by detection of water in the conduit 12 when the fixture is not activated, or within a set time period after activation. The message may not be transmitted unless a possible run-on condition is determined. The local computing device 129 may be a computing device, such as a server, a user device, a location smart display monitor, or any combination thereof, located on the same premises as the restroom 105. The transmitted message may be sent to the local computing device 129 via the bridge 125, and/or the transmitted message may be sent via bridge 125 and/or gateway 127 to the server 130. Additional information may be transmitted with the message. By transmitting the run-on detection message to the computing device, appropriate personnel can be alerted to the presence of a potential run-on condition in real-time. Any of the devices, including the detection system 30, the local computing device 129, the first user device 110, the second user device 115, and/or the server 130, may generate notifications, for example, if a restroom and/or fixture requires attention due to a run-on condition.

[0064] Bridge 125 may be configured to connect one or more fixtures via a network. The network may be a local area network, such as a building or corporate network. The bridge 125 may be a wired or wireless bridge. In preferred embodiments, the bridge 125 comprises a wireless interface to communicate (e.g., send/receive) with one or more fixtures. The wireless interface may use a short-range wireless communication protocol, such as Bluetooth® communications, Bluetooth® Low Energy communications, Wi-Fi communications, ANT communications, LoRa communications, Zig Bee Communications, or any equivalent thereof.

[0065] Gateway 127 may be configured to connect the network (e.g., building or corporate network) to a wide area network, such as network 150. The gateway 127 may provide interoperability between building or corporate network and network 150. The gateway 127 may comprise protocol translators, impedance matchers, rate converters, fault isolators, or signal translators. In some examples, the gateway 127 may perform protocol conversions to connect networks with different network protocol technologies.

[0066] First user device 110 may be a mobile device, such as a cellular phone, a mobile phone, a smart phone, a tablet, a laptop, or an equivalent thereof. First user device 110 may provide a first user with access to various applications and services. For example, first user device 110 may provide the first user with access to the Internet. Additionally, first user device 110 may provide the first user with one or more applications (“apps”) located thereon. The one or more

applications may provide the first user with a plurality of tools and access to a variety of services. In some embodiments, the one or more applications may include an application that provides access to a dashboard, or portal, that provides information about potential run-on detections. It is understood that the information provided may include additional information about the restroom 105 and/or critical diagnostics. Additionally or alternatively, the information may include information about individual fixtures, including, for example, real-time information about whether a fixture is currently being used. The application may comprise an authentication process to verify (e.g., authenticate) the identity of the first user prior to granting access to the dashboard (e.g. portal).

[0067] Second user device 115 may be a device configured to allow a user to execute software for a variety of purposes. Second user device 115 may belong to the first user that accesses first user device 110, or, alternatively, second user device 115 may belong to a second user, different from the first user. Second user device 115 may be a desktop computer, laptop computer, or, alternatively, a virtual computer. The software of second user device 115 may include one or more web browsers that provide access to websites on the Internet. These websites may include plumbing websites that allow the user to view information about a building's plumbing, an individual bathroom, and/or an individual fixture. In some embodiments, second user device 115 may include an application that allows the user to access a dashboard, or portal, to view information about a building's plumbing, an individual bathroom, and/or an individual fixture. The information may include any information disclosed herein. The website and/or the application may comprise an authentication component to verify (e.g., authenticate) the identity of the second user prior to granting access to the dashboard (e.g., portal).

[0068] Server 130 may be any server capable of executing application 132. Additionally, server 130 may be communicatively coupled to a database 140. In this regard, server 130 may be a stand-alone server, a corporate server, or a server located in a server farm or cloud-computer environment. According to some examples, server 130 may be a virtual server hosted on hardware capable of supporting a plurality of virtual servers. In some instances, the server 130 may be hosted by a commercial plumbing supply company, such as Sloan Valve Company. The server 130 may be hosted in a cloud provider, such as Microsoft Azure Cloud Service or an equivalent thereof. The server may execute application 132 on behalf of one or more consumers of the products manufactured and distributed by the commercial plumbing supply company.

[0069] The application 132 may be server-based software configured to provide users with information about restroom 105. In some embodiments, the application 132 may be server-based software that corresponds to client-based software executing on first user device 110 and/or second user device 115. Additionally, or alternatively, the application 132 may provide users access to the information through a website, or portal, accessed by first user device 110 or second user device 115 via network 150. The application 132 may comprise an authentication module to verify users before granting access to the information. The information may include a start time of the fixture's usage, an end time of the fixture's usage, a duration of the fixture's usage, etc. The application 132 may also analyze the information from

a plurality of fixtures associated with a location and present the analysis to a user, for example, via the dashboard. That is, the application 132 may receive information from each of a plurality of fixtures located in a restroom (e.g., restroom 105). The application 132 may then analyze the information associated with the fixture and present the analysis to a user, via the dashboard. The analysis may include determination of the existence of a run-on condition. It will be appreciated that the dashboard may allow a user to view information about the restroom or the building as-a-whole. The application 132 may also provide real-time information regarding whether a fixture is currently in use. Further, the dashboard may generate notifications, for example, if a restroom and/or fixture requires attention due to a run-on condition. The notifications may be an electronic communication, such as an email, a text message, a push notification, etc. Additionally or alternatively, the notifications may be displayed via an alert in the dashboard or location smart display monitor.

[0070] The database 140 may be configured to store information on behalf of application 132. The information may include, but is not limited to, data about restrooms, such as the quantity, type, model numbers, etc. of the fixtures associated with a restroom. Additionally or alternatively, the information stored in database 140 may comprise data regarding run-on conditions of each fixture. User-preferences may also be stored in the database 140. The user-preferences may define how users receive notifications, alerts, etc. The database 140 may include, but is not limited to relational databases, hierarchical databases, distributed databases, in-memory databases, flat file databases, XML databases, NoSQL databases, graph databases, and/or a combination thereof.

[0071] Network 150 may include any type of network. In this regard, first network 150 may include the Internet, a local area network (LAN), a wide area network (WAN), a wireless telecommunications network, and/or any other communication network or combination thereof. It will be appreciated that the network connections shown are illustrative and any means of establishing a communications link between the computers may be used. The existence of any of various network protocols such as TCP/IP, Ethernet, FTP, HTTP and the like, and of various wireless communication technologies such as GSM, CDMA, WiFi, and LTE, is presumed, and the various computing devices described herein may be configured to communicate using any of these network protocols or technologies. The data transferred to and from various computing devices in environment 100 may include secure and sensitive data, such as confidential documents, customer personally identifiable information, and account data. Therefore, it may be desirable to protect transmissions of such data using secure network protocols and encryption, and/or to protect the integrity of the data when stored on the various computing devices. For example, a file-based integration scheme or a service-based integration scheme may be utilized for transmitting data between the various computing devices. Data may be transmitted using various network communication protocols. Secure data transmission protocols and/or encryption may be used in file transfers to protect the integrity of the data, for example, File Transfer Protocol (FTP), Secure File Transfer Protocol (SFTP), and/or Pretty Good Privacy (PGP) encryption. In many embodiments, one or more web services may be implemented within the various computing devices. Web services may be accessed by authorized external devices and

users to support input, extraction, and manipulation of data between the various computing devices in the environment **100**. Web services built to support a personalized display system may be cross-domain and/or cross-platform, and may be built for enterprise use. Data may be transmitted using the Secure Sockets Layer (SSL) or Transport Layer Security (TLS) protocol to provide secure connections between the computing devices. Web services may be implemented using the WS-Security standard, providing for secure SOAP messages using XML encryption. Specialized hardware may be used to provide secure web services. For example, secure network appliances may include built-in features such as hardware-accelerated SSL and HTTPS, WS-Security, and/or firewalls. Such specialized hardware may be installed and configured in environment **100** in front of one or more computing devices such that any external devices may communicate directly with the specialized hardware.

[0072] In another embodiment, the detection system **30** may be present in an environment including a cloud service, connected via network **150**. Network **150** may be the same, or similar, to network **150** described above.

[0073] The detection system **30** may be Internet-of-Things (IoT)-enabled. IoT capability may enable the detection system **30** to be monitored and/or managed remotely from a central dashboard. While run-on may be detected by techniques such as manual examination, this requires service personnel to manually check each unit and/or user reporting, which may result in the run-on condition existing for a period of time before service personnel are aware of the need for service. The present disclosure describes using IoT connectivity to enable a potential run-on condition to be communicated to a central dashboard and notifications and alerts to be viewed and sent to relevant personnel as needed to ensure maintenance is performed in a timely manner.

[0074] Various embodiments of detection systems **30** and plumbing fixture assemblies **10** have been described herein, which include various components and features. In other embodiments, the system and assembly may be provided with any combination of such components and features. It is also understood that in other embodiments, the various devices, components, and features of the system and assembly described herein may be constructed with similar structural and functional elements having different configurations, including different ornamental appearances.

[0075] The detection system **30** and the plumbing fixture assembly **10** incorporating the system **30** provide advantages over existing technologies. For example, the use of the system **30** provides the ability to detect potential run-on and notify service personnel in real time, to permit rapid servicing. As another example, the use of a water contact sensor, effectively detecting a resistance change across electrical probes, provides a low cost, low power, robust, simple, effective means of detecting run-on, and also provides a low flow-resistance solution, thereby not negatively influencing the dynamic pressure within the conduit **12** and therefore flush volumes and fixture performance. Still other advantages are recognized to those skilled in the art.

[0076] Several alternative embodiments and examples have been described and illustrated herein. A person of ordinary skill in the art would appreciate the features of the individual embodiments, and the possible combinations and variations of the components. A person of ordinary skill in the art would further appreciate that any of the embodiments

could be provided in any combination with the other embodiments disclosed herein. It is understood that the invention may be embodied in other specific forms without departing from the spirit or central characteristics thereof. The present examples and embodiments, therefore, are to be considered in all respects as illustrative and not restrictive, and the invention is not to be limited to the details given herein. The terms “top,” “bottom,” and the like, as used herein, are intended for illustrative purposes only and do not limit the embodiments in any way. Nothing in this specification should be construed as requiring a specific three dimensional orientation of structures in order to fall within the scope of this invention, unless explicitly specified by the claims. When used in description of a method or process, the term “providing” (or variations thereof) as used herein means generally making an article available for further actions, and does not imply that the entity “providing” the article manufactured, assembled, or otherwise produced the article. Additionally, the term “plurality,” as used herein, indicates any number greater than one, either disjunctively or conjunctively, as necessary, up to an infinite number. Accordingly, while the specific embodiments have been illustrated and described, numerous modifications come to mind without significantly departing from the spirit of the invention and the scope of protection is only limited by the scope of the accompanying claims.

1. A system for detection of water in a conduit, comprising:

- a casing configured to be attached to the conduit on an exterior of the conduit;
- a sensor assembly supported by the casing and configured to sense a presence of water within the conduit, the sensor assembly comprising a first probe having a first probe contact accessible within the casing, the first probe configured to extend into an interior of the conduit to be exposed in the interior of the conduit, and a second probe having a second probe contact accessible within the casing, the second probe configured to extend into the interior of the conduit to be exposed in the interior of the conduit, wherein the first probe and the second probe are spaced from each other and electrically isolated from each other;

an electronic component contained within the casing and connected to the first probe contact and the second probe contact to place the electronic component in electronic communication with the first probe and the second probe.

2. The system of claim 1, wherein the casing comprises a main casing configured to be attached to the conduit and supporting the sensor assembly, and a secondary casing removably connected to the main casing, wherein the secondary casing contains the electronic component.

3. The system of claim 2, further comprising a first contact and a second contact supported by the secondary casing and electronically connected to the electronic component, wherein the first contact and the second contact are positioned to engage the first probe contact and the second probe contact when the secondary casing is connected to the main casing, thereby placing the first probe and the second probe in electronic communication with the electronic component.

4. The system of claim 2, wherein the main casing has a slot on an outer side thereof, and the secondary casing has a projection on an inner side thereof, and wherein the

secondary casing is removably connected to the main casing by the projection being received in the slot.

5. The system of claim 4, wherein the slot and the projection are oriented horizontally, and the secondary casing is removably connected to the main casing by the projection sliding horizontally into the slot.

6. The system of claim 2, wherein the main casing has a central passage configured to receive the conduit there-through, such that the main casing is configured to extend around the conduit, and the main casing comprises two pieces joined together, with each of the two pieces defining a portion of the central passage.

7. The system of claim 1, wherein the sensor assembly further comprises a sensor body having a first passage and a second passage therethrough, the second passage spaced from the first passage, the sensor body comprising an outer portion configured to be positioned on an exterior of the conduit and a projection configured to extend through a hole in the conduit and into an interior of the conduit, and wherein the first probe and the second probe are received through the first passage and the second passage, respectively, and configured to be exposed in the interior of the conduit.

8. The system of claim 7, wherein the outer portion is wider than the projection and configured to permit sealing of the outer portion around the hole.

9. The system of claim 8, wherein the casing has an inner wall configured to confront an exterior of the conduit and an opening in the inner wall receiving a portion of the outer portion of the sensor body, and wherein the inner wall of the casing is recessed around the opening to receive a lip of the outer portion of the sensor body, such that the lip of the sensor body is configured to be received between the inner wall of the casing and the exterior of the conduit.

10. The system of claim 1, wherein the electronic component comprises a power source configured to deliver power to the sensor assembly.

11. The system of claim 1, further comprising a plurality of electronic components, including the electronic component, contained within the casing, wherein the plurality of electronic components comprise a power source, a processor, a memory, and a communication device configured for communication with an external device.

12. An assembly comprising the system of claim 1 and the conduit, wherein the conduit is configured to allow passage of water between a valve and a plumbing fixture, further comprising a restrictor tray positioned within the interior of the conduit, adjacent to the first probe and the second probe, wherein the restrictor tray comprises a vessel configured to collect the water flowing through the conduit and to restrict a rate of the water exiting the vessel to a sufficient degree to permit the first probe and the second probe to contact the water during a low flow run-on condition.

13. A vacuum breaker assembly for use with a plumbing fixture and a valve configured to control discharge of water through an outlet and into the plumbing fixture, the vacuum breaker assembly comprising:

- a vacuum breaker tube configured to be connected in fluid communication with the outlet of the valve, between the valve and the plumbing fixture, the vacuum breaker tube having a wall and defining an interior to receive the discharge of the water from the outlet;
- a vacuum breaker sleeve received within the vacuum breaker tube and fixed in position within the vacuum

breaker tube, the vacuum breaker sleeve defining an inner passage, such that water discharged through the outlet is configured to flow through the vacuum breaker tube and the inner passage of the vacuum breaker sleeve to the plumbing fixture;

- a vacuum breaker supported by the vacuum breaker sleeve and positioned within the inner passage of the vacuum breaker sleeve, wherein the vacuum breaker is configured to prevent backflow of water from the plumbing fixture into the valve; and
- a sensor assembly configured to sense a presence of water within the vacuum breaker tube and comprising a first probe extending through the wall of the vacuum breaker tube and into the interior of the vacuum breaker tube, and a second probe extending through the wall of the vacuum breaker tube and into the interior of the vacuum breaker tube, wherein the first probe and the second probe are spaced from each other and electrically isolated from each other.

14. The vacuum breaker assembly of claim 13, further comprising a casing attached to the wall of the vacuum breaker tube and positioned on an exterior of the vacuum breaker tube, wherein the casing supports the first probe and the second probe of the sensor assembly.

15. The vacuum breaker assembly of claim 14, further comprising an electronic component contained within the casing and connected to the first probe and the second probe to place the electronic component in electronic communication with the first probe and the second probe.

16. The vacuum breaker assembly of claim 15, wherein the casing comprises a main casing attached to the vacuum breaker tube and supporting the sensor assembly, and a secondary casing removably connected to the main casing, wherein the secondary casing contains the electronic component.

17. The vacuum breaker assembly of claim 14, wherein the casing has a central passage configured to receive the vacuum breaker tube therethrough, such that the casing is configured to extend around the vacuum breaker tube.

18. The vacuum breaker assembly of claim 13, wherein the sensor assembly further comprises a sensor body having a first passage and a second passage therethrough, the second passage spaced from the first passage, the sensor body comprising an outer portion positioned on an exterior of the wall of the vacuum breaker tube and a projection extending through a hole in the wall of the vacuum breaker tube and into the interior of the vacuum breaker tube, and wherein the first probe and the second probe are received through the first passage and the second passage, respectively, and are exposed in the interior of the vacuum breaker tube.

19. The vacuum breaker assembly of claim 13, further comprising a restrictor tray positioned within the interior of the vacuum breaker tube, adjacent to the first probe and the second probe, wherein the restrictor tray comprises a vessel configured to collect the water flowing through the vacuum breaker tube and to restrict a rate of the water exiting the vessel to a sufficient degree to permit the first probe and the second probe to contact the water during a low flow run-on condition.

20. (canceled)

21. A sensor for detecting presence of water in a conduit, comprising:

- a sensor body having a first passage and a second passage therethrough, the second passage spaced from the first passage, the sensor body comprising an outer portion configured to be positioned on an exterior of the conduit and a projection configured to extend through a hole in the conduit and into an interior of the conduit, wherein the outer portion is wider than the projection to permit sealing of the outer portion around the hole, the sensor body comprising an insulative material located between the first passage and the second passage;
 - a first probe received in the first passage and having a first probe contact accessible at the outer portion of the sensor body, the first probe extending through the first passage and configured to be exposed in the interior of the conduit; and
 - a second probe received in the second passage and having a second probe contact accessible at the outer portion of the sensor body, the second probe extending through the second passage and configured to be exposed in the interior of the conduit,
- wherein the insulative material of the sensor body electrically isolates the first probe from the second probe when the first and second probes are received in the first and second passages, respectively.

22-33. (canceled)

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